

Data Science and Big Data (MCS-226) *(Job/Research-based Course)*

[Block-1]

By

Dr. Sumit Tetarave

Research Direction and Assignment:

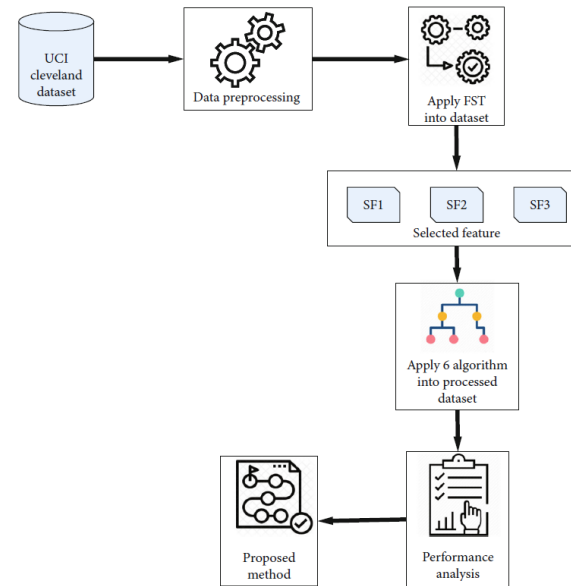
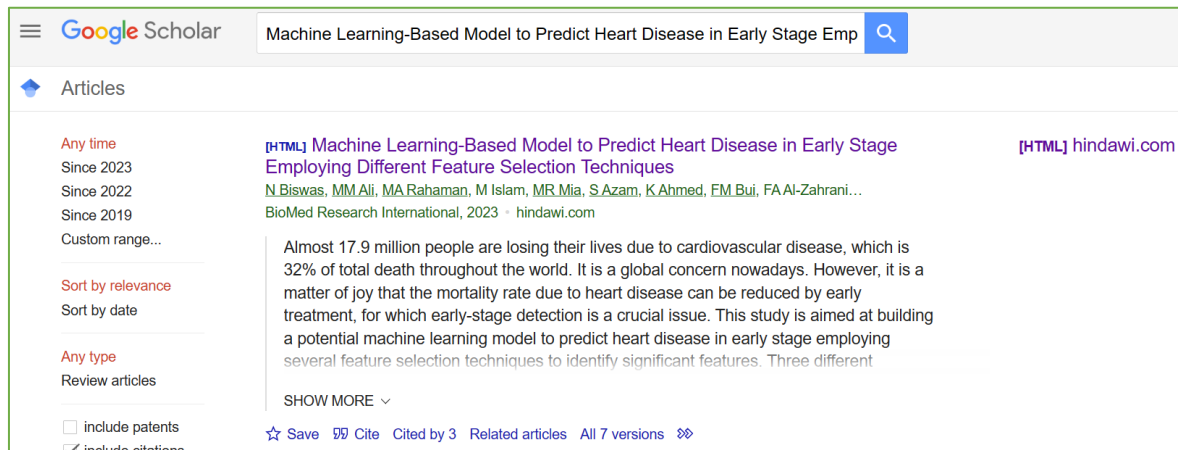


FIGURE 1: Workflow of predicting heart disease.

TABLE 3: Feature score using FST1.				
Order	Feature	Feature name	Code	Scores
1	9	exang	EIA	70.95
2	3	cp	CPT	69.77
3	10	Oldpeak	OP	68.55
4	8	Thalach	MHR	65.12
5	12	ca	NMV	64.05
6	11	Slope	PES	40.90
7	13	thal	TS	31.80
8	2	Sex	SEX	25.79
9	1	Age	AGE	16.12
10	4	trestbps	RBP	6.46
11	7	restecg	REC	5.78
12	5	chol	CM	2.20
13	6	lbs	FBS	0.24

Research Article

Machine Learning-Based Model to Predict Heart Disease in Early Stage Employing Different Feature Selection Techniques

Niloy Biswas¹, Md Mamun Ali¹, Md Abdur Rahaman¹, Minhajul Islam¹, Md. Rajib Mia¹, Sami Azam², Kawsar Ahmed^{3,4}, Francis M. Bui⁴, Fahad Ahmed Al-Zahrani⁵, and Mohammad Ali Moni⁶

Almost 17.9 million people are losing their lives due to cardiovascular disease, which is 32% of total death throughout the world. It is a global concern nowadays. However, it is a matter of joy that the mortality rate due to heart disease can be reduced by early treatment, for which early-stage detection is a crucial issue. This study is aimed at building a potential machine learning model to predict heart disease in early stage employing several feature selection techniques to identify significant features. Three different approaches were applied for feature selection such as chi-square, ANOVA, and mutual information, and the selected feature subsets were denoted as SF1, SF2, and SF3, respectively. Then, six different machine learning models such as logistic regression (C1), support vector machine (C2), K-nearest neighbor (C3), random forest (C4), Naive Bayes (C5), and decision tree (C6) were applied to find the most optimistic model along with the best-fit feature subset. Finally, we found that random forest provided the most optimistic performance for SF3 feature subsets with 94.51% accuracy, 94.87% sensitivity, 94.23% specificity, 94.95 area under ROC curve (AURC), and 0.31 log loss. The performance of the applied model along with selected features indicates that the proposed model is highly potential for clinical use to predict heart disease in the early stages with low cost and less time.

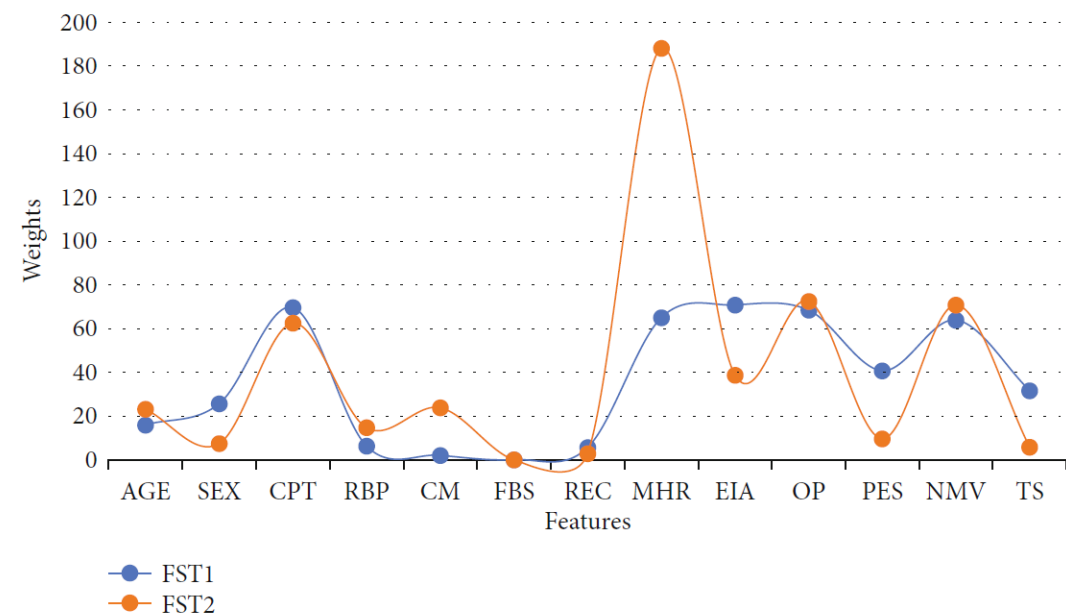


FIGURE 2: Feature score by FST1 and FST2.

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This page displays an alphabetical list of all the databases on PhysioNet. To s resource type if needed, and select how you would like the results to be orde

Each project is made available under one of the following access policies:

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AF Termination Challenge Database: ECG recordings created for the Co termination of atrial fibrillation.

AHA Database Sample Excluded Record: Two ECG signals that were exc

A large scale 12-lead electrocardiogram database for arrhythmia study: 10,000 patients

A multi-camera and multimodal dataset for posture and gait analysis: A used in gait and posture rehabilitation. It is equipped with a pair of Def participant.

ANSI/AAMI EC13 Test Waveforms: The files in this set can be used for t both synthetic and real waveforms.

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About 3,14,000 results (0.23 sec)

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ZN Mohammed, SM Kadhem - Engineering and Technology Journal, 2021 - iasj.net

... make a purchase or not from their previous behaviors. ... the main aim of this study is to

[PDF] iasj.net

REFERENCES

[1] write research paper name

[2] write 2nd paper name

[3]

Now, read the research papers and fill the following table1. Table 1 shows a review of current research to understand the ongoing works on your downloaded datasets.

Paper Reference	Purpose of the paper	Methodology with accuracy	Is AI/ML used? If yes, write name.	Details of their Dataset, such as size, attributes, null values in each attributes	Accuracy with your downloaded dataset.
[1]					

Medica Project 1: Prediction of Sepsis

<https://github.com/ShwetaMaharanawar/Accurate-Prediction-of-Sepsis-in-ICU-Patients/tree/my-new-branch>

The screenshot shows the GitHub interface for the repository 'ShwetaMaharanawar / Accurate-Prediction-of-Sepsis-in-ICU-Patients'. The current branch is 'my-new-branch'. The repository is public and has 2 branches and 0 tags. The file list includes folders 'Certificate', 'Output', 'research papers', and 'sepsis', and files 'Group 16 Accurate Prediction of Sepsis in ICU ...' and 'Project Log Book_Group_16.docx'. A 'Clone' dropdown menu is open, showing options: 'Clone' (with a question mark), 'HTTPS' (selected), 'GitHub CLI', and 'Open with GitHub Desktop'. The 'Download ZIP' option is circled in red with an arrow pointing to it. The 'About' section on the right describes the project: 'Developed an early sepsis detection model using Random Forest and SOFA/qSOFA scores, & launched website to raise awareness & provide healthcare professionals access.' It also shows '0 stars', '1 watching', and '0 forks'.

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Go to file

Clone ?

HTTPS GitHub CLI

https://github.com/ShwetaMaharanawar/Accurate-P... Clone using the web URL.

Open with GitHub Desktop

Download ZIP

2 weeks ago

About

Developed an early sepsis detection model using Random Forest and SOFA/qSOFA scores, & launched website to raise awareness & provide healthcare professionals access.

Readme Activity 0 stars 1 watching 0 forks

Download Dataset

<https://www.kaggle.com/datasets/salikhussaini49/prediction-of-sepsis/code>

The screenshot shows the Kaggle interface for the 'Prediction of Sepsis' dataset. The left sidebar contains navigation links: Home, Competitions, Datasets (highlighted), Models, Code, Discussions, Learn, More, Your Work, and a list of viewed datasets including 'Prediction of Sepsis'. The main content area displays the dataset title 'Prediction of Sepsis' with tabs for Data Card, Code (9), Discussion (0), and Suggestions (0). Below the tabs, a list of features is shown: Temp Temperature (Deg C), SBP Systolic BP (mm Hg), MAP Mean arterial pressure (mm Hg), DBP Diastolic BP (mm Hg), and Resp Respiration rate (breaths per minute). A 'View more' link is present. The 'Dataset.csv' file (153.82 MB) is highlighted, and its download icon is circled in red with an arrow. The 'Data Explorer' on the right shows the file structure, with 'Dataset.csv' also circled in red. The browser address bar shows the URL: https://www.kaggle.com/datasets/salikhussaini49/prediction-of-sepsis/data.

← ↻ 🔍 <https://www.kaggle.com/datasets/salikhussaini49/prediction-of-sepsis/data>

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Data Card Code (9) Discussion (0) Suggestions (0)

Temp Temperature (Deg C)

SBP Systolic BP (mm Hg)

MAP Mean arterial pressure (mm Hg)

DBP Diastolic BP (mm Hg)

Resp Respiration rate (breaths per minute)

View more

Dataset.csv (153.82 MB)

Detail Compact Column

10 of 44 columns

About this file

This file does not have a description yet.

Add Suggestion

Data Explorer

Version 2 (410.81 MB)

- training_setA
- training_setB
- Dataset.csv**
- LICENSE.txt
- SHA256SUMS.txt
- physionet_challenge_2019
- utility_nonsepsis_diagram
- utility_sepsis_diagram.svg

Step 1: Open your Google Colab and Upload the python files

The screenshot shows the Google Colab web interface in a browser. The browser's address bar displays `https://colab.research.google.com`. The main header of the Colab interface says "Welcome To Colab" with a menu bar containing "File", "Edit", "View", and "Insert". On the left, there is a "Table of contents" sidebar with links to "Getting started", "Data science", "Machine learning", and "More Resources", along with a "+ Section" button. A central "Open notebook" dialog box is open, featuring a list on the left with "Examples", "Recent", "Google Drive", "GitHub", and "Upload" (which is highlighted). The main area of the dialog shows a large cloud icon, a blue "Browse" button, and the text "or drag a file here". At the bottom of the dialog are two buttons: "+ New notebook" and "Cancel". The browser's taskbar at the bottom shows various application icons, including the Windows Start button, search bar, and system tray with the time 10:59 AM and date 06-11-2024.

You can find all python files in your Google My Drive-> Colab Notebooks.
Step 2: Now upload your dataset (Dataset.csv) here.

The screenshot shows the Google Drive web interface. On the left is a sidebar with navigation options: Home, Activity, Workspaces, My Drive, Computers, Shared with me, and Recent. The main area shows the 'My Drive' > 'Colab Notebooks' path. Below the path are filters for Type, People, and Modified. A list of files is displayed with headers 'Name' and an upward arrow. The files are: 'Copy of Welcome To Colab' (Colab icon), 'Step1_Project2.ipynb' (Colab icon), and 'UNSW-NB15_2.csv' (CSV icon). Each file has a share icon to its right.

Drive

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+ New

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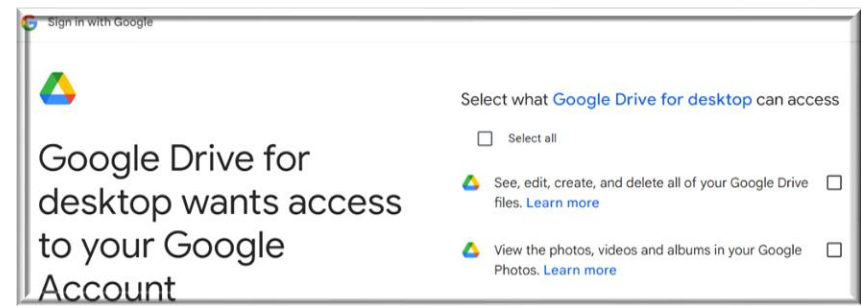
My Drive > Colab Notebooks

Type People Modified

Name ↑

- Copy of Welcome To Colab
- Step1_Project2.ipynb
- UNSW-NB15_2.csv

Step 3: While running/mounting [3], Google wants permission.
Give the permission.



```
from google.colab import drive
drive.mount('/content/drive')
```

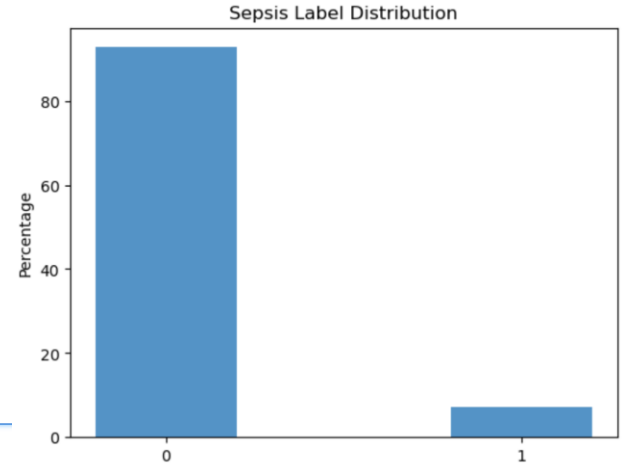
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive",

```
import pandas as pd
dataSepsis = pd.read_csv("/content/drive/MyDrive/Colab Notebooks/sepsis/Dataset.csv", sep=",")
```

```
dataSepsis.head(5)
```

	Unnamed: 0	Hour	HR	O2Sat	Temp	SBP	MAP	DBP	Resp	EtCO2	...	Fibrinogen	Platelets	Age	Gender
0	0	0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	68.54	0
1	1	1	65.0	100.0	NaN	NaN	72.0	NaN	16.5	NaN	...	NaN	NaN	68.54	0

Used Data Pre-processing Method-1



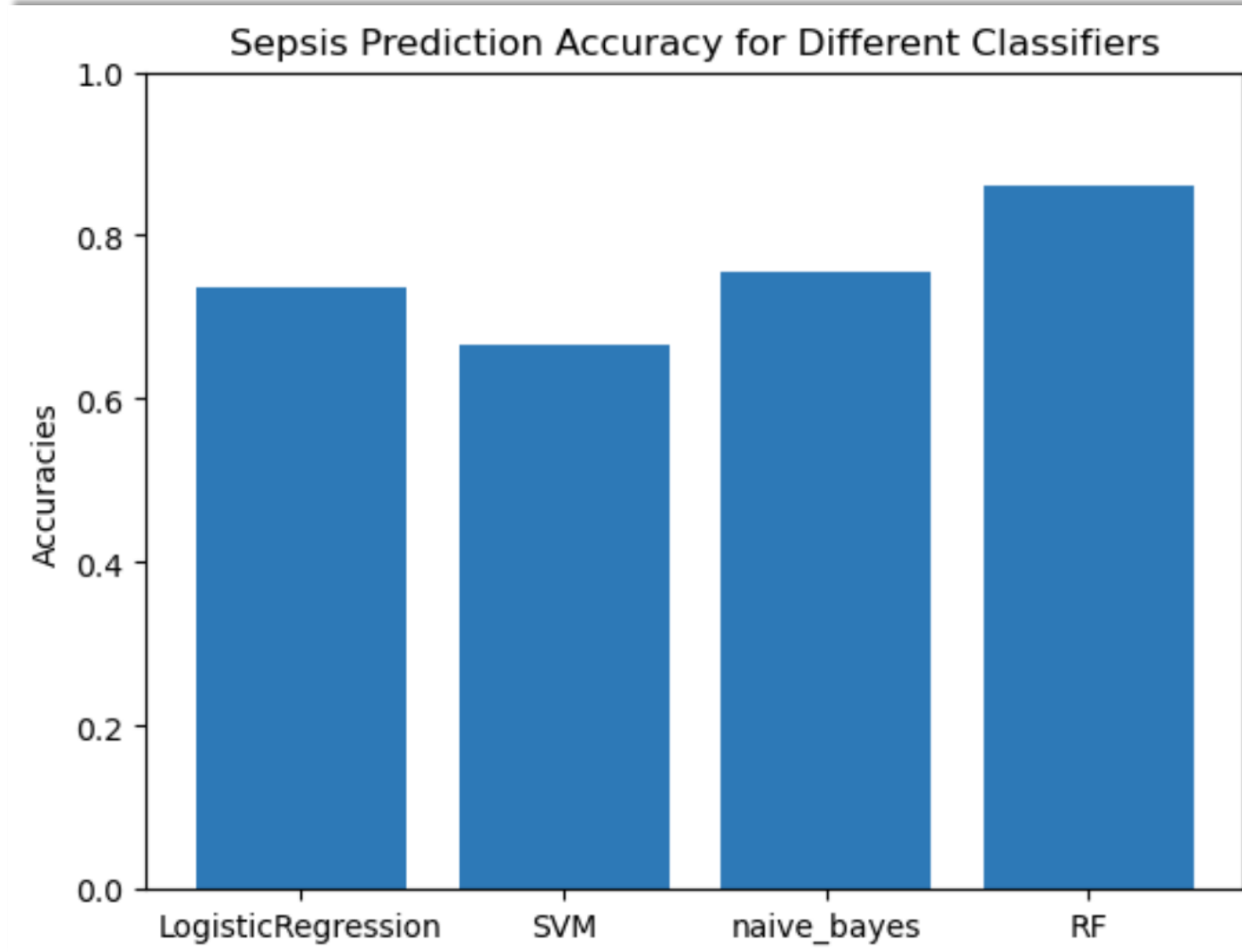
```
[12]: Sepsis_counts = data['SepsisLabel_max'].value_counts()
# Explanation: In this line, you are counting the number of instances for each unique value in the
# "SepsisLabel_max" column of the "data" DataFrame. This is a way to determine the distribution of sepsis
# labels in your updated dataset, which now includes a balanced number of instances from the majority and
# minority classes.

Sepsis_percentages = Sepsis_counts / len(data) * 100
# Explanation: Similar to previous lines, here you are calculating the percentages of each sepsis label in
# your dataset. You divide the counts in "Sepsis_counts" by the total number of instances in the "data"
# DataFrame and multiply by 100 to express the result as a percentage.

print(Sepsis_percentages)
# Explanation: This line prints the percentages of each sepsis label to the console.
# These percentages reflect the distribution of sepsis labels in your updated dataset, which you balanced
# by oversampling the minority class. Balancing the dataset can be important for training machine learning models,
# as it prevents one class from dominating the training process.
```

```
SepsisLabel_max
0    66.666667
1    33.333333
Name: count, dtype: float64
```

Used ML Models: **LOGISTIC REGRESSION, NAIVE BAYES**
and **SUPPORT VECTOR MACHINE, RANDOM FOREST**



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[Block-1]

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Recall from 2nd Sem Block 4: Classification, Clustering and Web Mining

- **Unit 10** covered:

- Overview of Classification
- **General approaches to solve a classification problem**
- Details of various classification techniques

Data classification is a **two-step process**, consisting of:

- 1) A learning step (or training phase)
 - where a classification model is **constructed**
- 2) A classification step
 - where the model is used to **predict** class labels for given data.

In the Step 1):

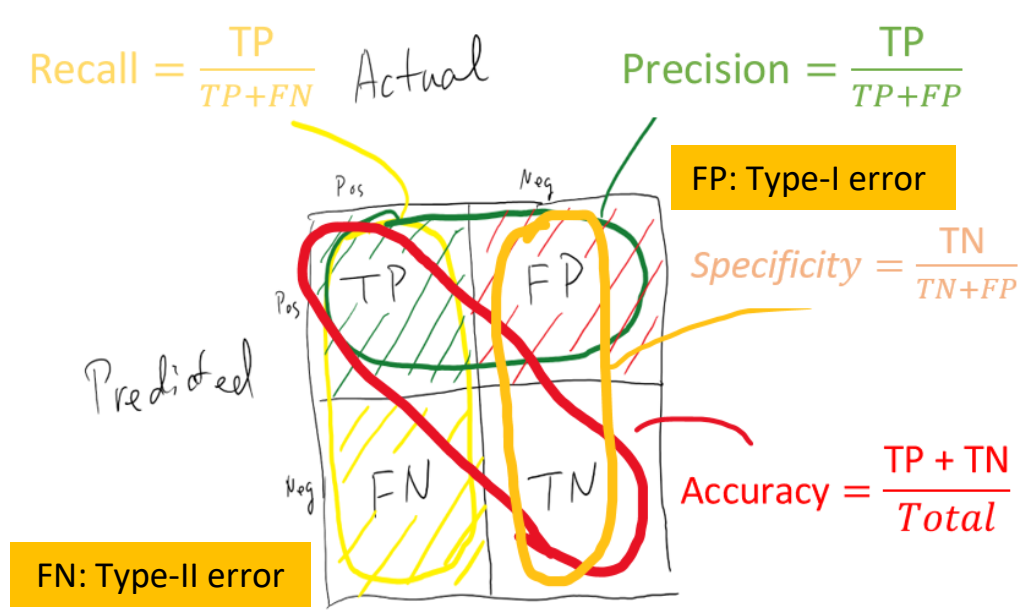
- Because the class label of each training tuple is provided, this step is also known as **supervised learning**
 - i.e., the learning of the classifier is “supervised” in that it is told to which class each training tuple belongs.
- It contrasts with unsupervised learning (or **clustering**), in which
 - the class label of each training tuple is **not known**, and
 - the number of classes to be learned may not be known in advance.

In the Step 2):

- A test set is used, made up of test tuples and their associated class labels.
 - They are independent of the training tuples,
 - ✓ meaning that they were not used to construct the classifier.
- The accuracy of a classifier on a given test set is the percentage of test set tuples that are correctly classified by the classifier.

Confusion Matrix: Performance measurement in classification problems

- **Accuracy:** From all the classes (positive and negative), how many of them we have predicted correctly.
(Accuracy should be high as possible)
- **Recall:** From all the positive classes, how many we predicted correctly.
(Recall should be high as possible)
- **Precision:** From all the classes we have predicted as positive, how many are actually positive.
(Precision should be high as possible)



y	y pred	output for threshold 0.6	Recall	Precision	Accuracy
0	0.5	0 TN	$= \frac{2}{2+2}$ 1/2	$= \frac{2}{2+1}$ 2/3	$= \frac{2+2}{7}$ 4/7
1	0.9	1 TP			
0	0.7	1 FP			
1	0.7	1 TP			
1	0.3	0 FN			
0	0.4	0 TN			
1	0.5	0 FN			

AUROC Curve (Area Under the Receiver Operating Characteristics) for classification problem

- TPR (True Positive Rate) / Recall / **Sensitivity**

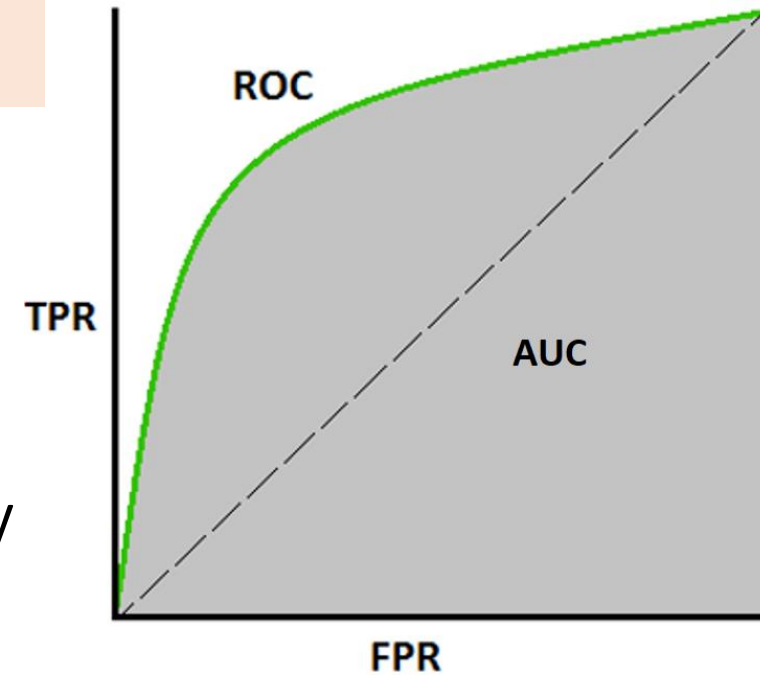
$$\text{TRP/ Recall / Sensitivity} = \frac{TP}{TP+FN}$$

- For example: Proportion of truly diseased persons in a screened population who are identified as being diseased by the test.
- It is the percentage of persons with the disease who are **correctly identified**

- **Specificity-**

- It is the percentage of persons **without the disease**
 - who are **correctly excluded** by the test.
- It is the proportion of truly non-diseased persons
 - who are so identified by the screening test.

- FPR (False Positive Rate)



$$\text{Specificity} = \frac{TN}{TN+FP}$$

$$\text{FPR} = 1 - \text{Specificity} = \frac{FP}{TN + FP}$$

Data Science and Big Data (MCS-226) *(Job/Research-based Course)*

[Block-1]

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Learning Process

1. Basic Methods of Data Analysis (Block-1)
 2. Data Preprocessing Methods and Visualization (Block-1)
 3. Architectures to handle Big Data (Block-2)
 4. Mining Techniques in Big Data (Block-3)
- ✓ ML methods (Block-4) will be covered in between.

UNIT 1: INTRODUCTION TO DATA SCIENCE

- **Data Science - Definition**

- **Types of Data**

- Statistical Data Types
- Sampling
- Basic Methods of Data Analysis
- Descriptive Analysis
- Exploratory Analysis
- Inferential Analysis
- Predictive Analysis
- Common Misconceptions of Data Analysis
- Applications of Data Science
- Data Science Life cycle

- UNIT 2: PORTABILITY AND STATISTICS FOR DATA SCIENCE

- UNIT 3: DATA PREPARATION FOR ANALYSIS

- UNIT 4: DATA VISUALIZATION AND INTERPRETATION

- Data Science is a multi-disciplinary science (Fig. 1) with an objective to perform data analysis
 - to generate knowledge that can be used for decision making.

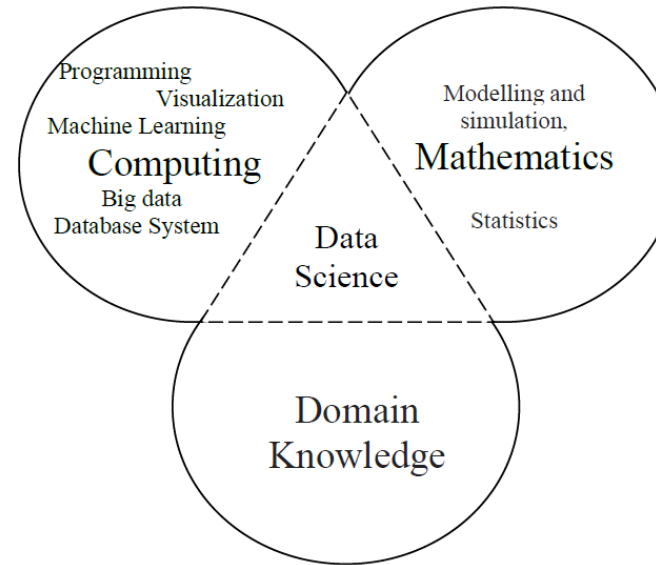


Fig. 1: Data Science

```
"Book": {  
  "Title": "Data Science",  
  "Price": 5000,  
  "Year": 2020  
}
```

Fig. 2: Semi-structured

TYPES OF DATA:

1. **Structured Data:** RDBMS allows structured storage.
2. **Semi-Structured Data:** The common form of semi-structured data is produced through XML, JSON objects, etc. (Fig. 2).
3. **Unstructured data:** The unstructured data does not follow any schema definition.
 - Unprocessed audio and video data.
4. **Data Streams:** A data stream is characterized by a sequence of data over a period of time.
 - Such data may be structured, semi-structured or unstructured, but it gets generated repeatedly.

UNIT 1: INTRODUCTION TO DATA SCIENCE

- Data Science - Definition
- Types of Data
- **Statistical Data Types**
- Sampling

- There are two distinct types of data that can be used in statistical analysis.
 - Quantitative data and Categorical data.

Quantitative (or Numerical) variables represent amounts of things

- For example- the number of trees in a forest.
 - ✓ Types of **quantitative variables** include:
 - Continuous** (aka **ratio variables**): represent measures and can usually be divided into units smaller than one (e.g. 0.75 grams).
 - Discrete** (aka **integer variables**): represent counts and usually can't be divided into units smaller than one (e.g. 1 tree).

Categorical variables represent groupings of things

- For example- the **different tree species** in a forest.
 - ✓ Types of categorical variables include:
 - Ordinal**: represent data with an order (e.g. rankings).
 - Nominal**: represent group names (e.g. brands or species names).
 - Binary**: represent data with a yes/no or 1/0 outcome (e.g. win or lose).

Understanding Measurement scale of data:

The following four characteristics to measure any scale of data:

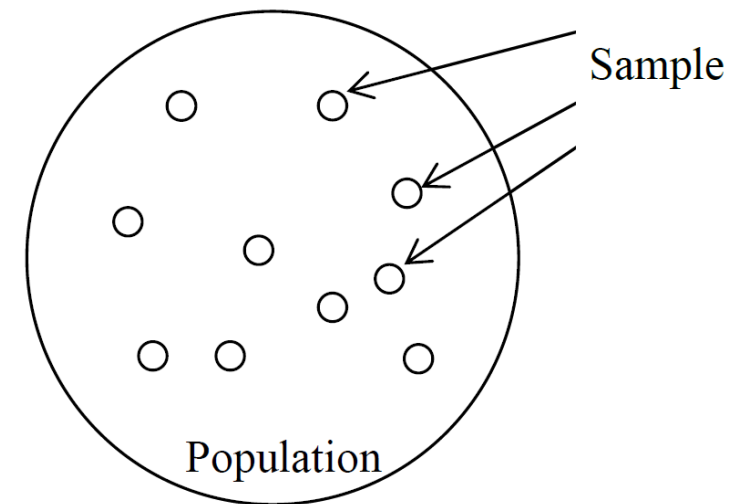
- Every representation of the measure should be unique, this is referred to as **identify of a value** (*IDV*).
- The second characteristics is the **magnitude** (*M*), which clearly can be used to compare the values,
 - for example, a weight of 70.5 kg is more than 70.2 kg.
- Third characteristics is about **equality in intervals** (*EI*) used to represent the data,
 - for example, the difference between 25 and 30 is 5 intervals, which is same as the difference between 41 to 46, which are also 5 intervals.
- The final characteristics is about a defined **minimum or zero value** (*MZV*),
 - for example, in Kelvin scale, temperature have an absolute zero value, whereas, the Intelligent quotient cannot be defined as zero.

Measurement Scale	Characteristics				Example
	<i>IDV</i>	<i>M</i>	<i>EI</i>	<i>MZV</i>	
Nominal	Yes	No	No	No	Gender F - Female M - Male
Ordinal	Yes	For rank ordering	No	No	A hypothetical Income Category: 1 - "0 or more but less than 26" 2 - "26 or more but less than 46" 3 - "46 or more but less than 61" 4 - "More than 61"
Interval	Yes	Yes	Yes	No	IQ (Intelligent quotient)- Level
Ratio	Yes	Yes	Yes	Yes	Temperature in K, Age

Sampling:

- In general, the size of data that is to be processed today is quite large.
- One of the key objectives of statistics, which uses sample data, is
 - to determine the statistic of the sample and
 - find the probability that the statistic developed for the sample
 - ✓ would determine the parameters of population with a specific percentage of accuracy.
- A good sample is representative of its population.

Term	Used for	Example
Statistic	Statistic is computed for the Sample	Sample mean ($\bar{x}$), Sample Standard deviation (s), Sample size (n)
Parameter	Parameters are predicted from sample and are about the Population	Population mean (μ), Population Std. deviation (σ), Population size (N)



UNIT 1: INTRODUCTION TO DATA SCIENCE

- Data Science - Definition
- Types of Data
- Statistical Data Types
- Sampling
- **Basic Methods of Data Analysis**

Some of Basic Methods of Data Analysis:

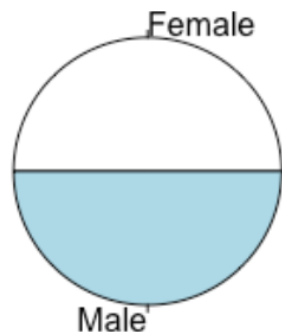
- Descriptive Analysis
- Exploratory Analysis
- Inferential Analysis
- Predictive Analysis

- **Descriptive Analysis**- is used to present basic summaries about data .
 - However, it makes no attempt to interpret the data. (i.e., **Uni-variate** analysis only)
- **Exploratory Analysis**- Instead of performing the final analysis,
 - we may like to explore the data for possible relationships using exploratory data analysis (using **Uni/bi/multi-variate** analysis), but
 - does not promise that a causal relationship may exist amongst variables.
- **Inferential Analysis**- Inferential analysis is performed to answer the question that
 - what is the probability of that the results obtained from an analysis can be applied to the entire population.
- **Predictive Analysis**- It aims to take predictions one step forward and
 - suggest solutions to present and future issues.

1. Descriptive analysis

For example- Sample Height Data

Enrolment Number	Gender	Height
S20200001	F	155
S20200002	F	160
S20200003	M	179
S20200004	F	175
S20200005	M	173
S20200006	M	160
S20200007	M	180
S20200008	F	178
S20200009	F	167
S20200010	M	173



Descriptive of Categorical Data:

Using by:

- Frequency Table or Pie Chart

Gender	Frequency	Proportion	Percentage
Female (F)	5	0.5	50%
Male (M)	5	0.5	50%

Descriptive of Quantitative Data:

- The height is a quantitative variable.
- The descriptive of quantitative data is given by the following two ways:
 1. Describing the central tendencies of the data
 2. Describing the spread of the data (or Dispersion or Variability).

1. Measures of Central tendency (Mean)

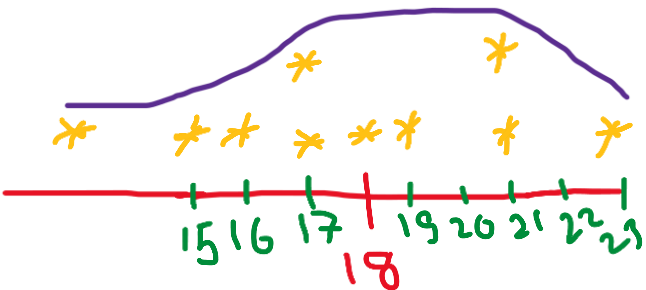
A Measure of Central Tendency is a one number summary of the data that typically describes the centre of the data. This one number summary is of three types.

- a. **Mean** : Mean is defined as the ratio of the sum of all the observations in the data to the total number of observations. This is also known as Average. Thus mean is a number around which the entire data set is spread.

Consider the following data points.

17, 16, 21, 18, 15, 17, 21, 19, 11, 23

$$\text{Mean} = \frac{17 + 16 + 21 + 18 + 15 + 17 + 21 + 19 + 11 + 23}{10} = \frac{178}{10} = 17.8$$

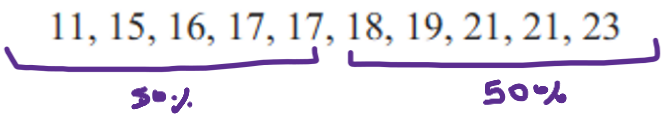


b. **Median** : Median is the point which divides the entire data into two equal halves. One-half of the data is less than the median, and the other half is greater than the same. Median is calculated by first arranging the data in either ascending or descending order.

- If the number of observations is odd, the median is given by the middle observation in the sorted form.
- If the number of observations are even, median is given by the mean of the two middle observations in the sorted form.

An important point to note is that the order of the data (ascending or descending) does not affect the median.

To calculate Median, let's arrange the data in ascending order.



Since the number of observations is even (10), median is given by the average of the two middle observations (5th and 6th here).

$$Median = \frac{5^{th} \text{ Obs} + 6^{th} \text{ Obs}}{2} = \frac{17 + 18}{2} = 17.5$$

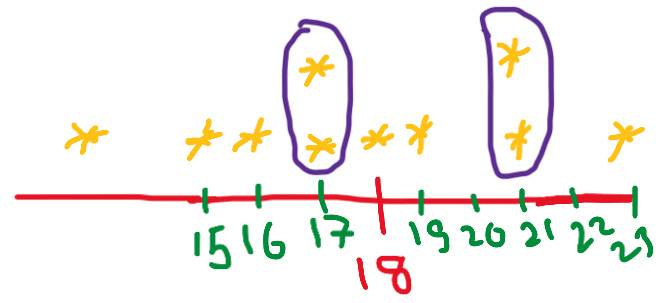
c. **Mode** : Mode is the number which has the maximum frequency in the entire data set, or in other words, mode is the number that appears the maximum number of times. A data can have one or more than one mode.

- If there is only one number that appears the maximum number of times, the data has one mode, and is called Uni-modal.
- If there are two numbers that appear the maximum number of times, the data has two modes, and is called Bi-modal.
- If there are more than two numbers that appear the maximum number of times, the data has more than two modes, and is called Multi-modal.

Consider the following data points.

17, 16, 21, 18, 15, 17, 21, 19, 11, 23

Mode is given by the number that occurs the maximum number of times. Here, 17 and 21 both occur twice. Hence, this is a Bimodal data and the modes are 17 and 21.



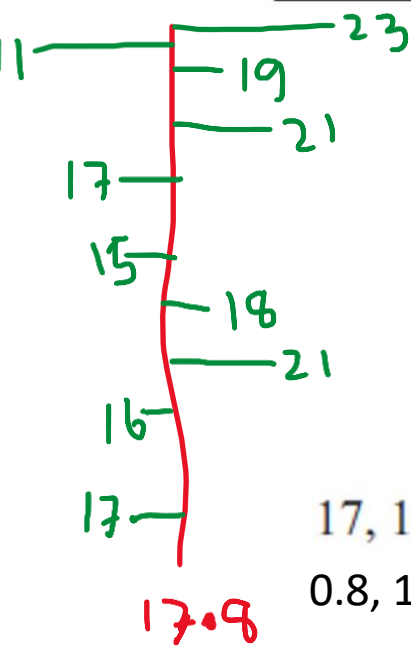
Mean is the accurate measure to describe the data when we do not have any outliers present. Median is used if there is an outlier in the dataset. Mode is used if there is an outlier AND about 1/2 or more of the data is the same.

2) Measures of Dispersion (or Variability)

Measures of Dispersion describes the spread of the data around the central value (or the Measures of Central Tendency)

1. **Absolute Deviation from Mean** — The Absolute Deviation from Mean, also called Mean Absolute Deviation (MAD), describes the variation in the data set, in the sense that it tells the average absolute distance of each data point in the set. It is calculated as

$$\text{Mean Absolute Deviation} = \frac{1}{N} \sum_{i=1}^N |X_i - \bar{X}|$$



17, 16, 21, 18, 15, 17, 21, 19, 11, 23
0.8, 1.8, 3.2, 0.2, 2.8, 3.2, 1.2, 6.8, 5.2
 $= 25.2/10 = 2.52$

2. **Variance** — Variance measures how far are data points spread out from the mean. A high variance indicates that data points are spread widely and a small variance indicates that the data points are closer to the mean of the data set. It is calculated as

$$\text{Variance} = \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2$$

3. **Standard Deviation** — The square root of Variance is called the Standard Deviation. It is calculated as

$$\text{Std Deviation} = \sqrt{\text{Variance}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2}$$

‘Mean’ is the only measure of central tendency that is affected by the outliers which in turn impacts Standard deviation.

with outlier	without outlier
Mean: 20.08	Mean: 12.72
Median: 14.0	Median: 13.0
Mode: 15	Mode: 15
Variance: 614.74	Variance: 21.28
Std dev: 24.79	Std dev: 4.61

[15, 101, 18, 7, 13, 16, 11, 21, 5, 15, 10, 9]
one can quickly say ‘101’ is an outlier
Mean is more affected than the Median.

4. **Range** — Range is the difference between the Maximum value and the Minimum value in the data set. It is given as

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

5. **Quartiles** — Quartiles are the points in the data set that divides the data set into four equal parts. Q1, Q2 and Q3 are the first, second and third quartile of the data set.

- 25% of the data points lie below Q1 and 75% lie above it.
- 50% of the data points lie below Q2 and 50% lie above it.
Q2 is nothing but Median.
- 75% of the data points lie below Q3 and 25% lie above it.



The IQR can also be used to identify suspected outliers. In general, a suspected outlier can exist in the following two ranges:

$$\begin{aligned}\text{Observation/values less than } Q1 - 1.5 \times \text{IQR} &= 4 - 16.5 = -12.5 \\ \text{Observation/values more than } Q3 + 1.5 \times \text{IQR} &= 15 + 16.5 = 31.5\end{aligned}$$

4, 17, 7, 14, 18, 12, 3, 16, 10, 4, 4, 11

Put them in order:

3, 4, 4, 4, 7, 10, 11, 12, 14, 16, 17, 18

Cut it into quarters:

3, 4, 4 | 4, 7, 10 | 11, 12, 14 | 16, 17, 18

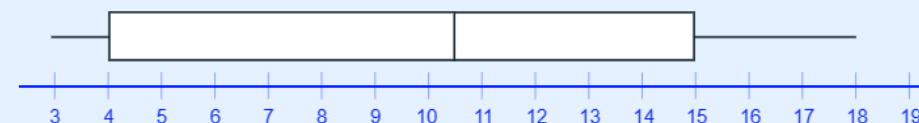
In this case all the quartiles are between numbers:

- Quartile 1 (Q1) = $(4+4)/2 = 4$
- Quartile 2 (Q2) = $(10+11)/2 = 10.5$
- Quartile 3 (Q3) = $(14+16)/2 = 15$

Also:

- The Lowest Value is **3**,
- The Highest Value is **18**

So now we have enough data for the **Box and Whisker Plot**:



And the **Interquartile Range** is:

$$Q3 - Q1 = 15 - 4 = 11$$

6. **Skewness** — The measure of asymmetry in a probability distribution is defined by Skewness. It can either be positive, negative or undefined.

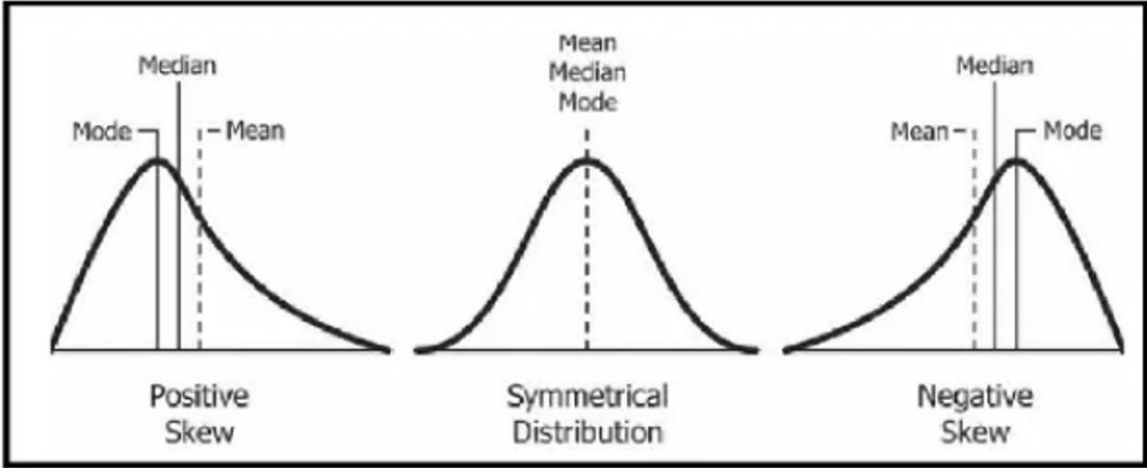
$$Skewness = \frac{3 (Mean - Median)}{Std\ Deviation}$$

The most commonly used method of calculating Skewness is

If the skewness is zero, the distribution is symmetrical. If it is negative, the distribution is Negatively Skewed and if it is positive, it is Positively Skewed.

Positive Skew — This is the case when the tail on the right side of the curve is bigger than that on the left side. For these distributions, mean is greater than the mode.

Negative Skew — This is the case when the tail on the left side of the curve is bigger than that on the right side. For these distributions, mean is smaller than the mode.



Explanatory or Independent Variables				Response or Dependent Variables
House Age	House Facing Direction	Distance from Station	Rent	Type
Two Years	South	15 KM	30,000	Elite
Five Years	North	10 KM	15,000	Non- elite
Ten Years	East	1 KM	25,000	Non- elite

A sample dataset

2. Exploratory Analysis

Some of the common methods that we can perform during exploratory analysis are as follows:

1. As a first step, we may perform the **descriptive of various categorical and qualitative** variables of our data.
 - Such information is very useful in determining the suitability of data for the purpose of analysis.
 - This may also help us in data cleaning, modification and transformation of data.
- a) For the qualitative data,
 - ✓ A **balanced distribution of data** among categories is most desirable.
 - ✓ However, such distribution may not be possible in actual situations.
 - ✓ Use **frequency tables** and **bar/pie charts** to know the distribution
- b) For the quantitative data,
 - ✓ We may compute the **mean, median, standard deviation, skewness** and **kurtosis**.
 - ✓ The kurtosis value relates to peaks (determined by mode) in the data.
 - ✓ Use **distribution/probability plot** or **histogram** to look into frequency distribution.

2. Next, after performing the univariate analysis,

- We may try to perform some **bi-variate** analysis.
- ✓ Some of the basic statistics that we can perform for bivariate analysis includes the following:
 - a) Make **two-way table** between categorical variables and make related **stacked bar charts**. We may also use **chi-square testing** find any significant relationships.
 - b) We may draw **side-by-side box plots** to check if the data of various categories have differences.
 - c) We may draw **scatterplot** and check the **correlation coefficient**, if that exists between two variables.

3. Finally, we may like to look into the possibilities of **multi-variate** relationships amongst data.

- We may use **dimensionality reduction** by using techniques
 - Feature Extraction or Principle Component Analysis,
- We may perform **clustering** to identify possible set of classes in the solution space or
- We may use graphical tools, like **bubble charts**, to visualize the data.

Example- Exploratory Analysis: Consider the previous sample data of students given about Gender and Height.

- ✓ Let us explore this data for an analytical question: **Does Height depends on Gender?**

We can perform the exploratory analysis on this data by drawing a side-by-side box plot for Male and Female students' height. This box plot is shown in Figure.

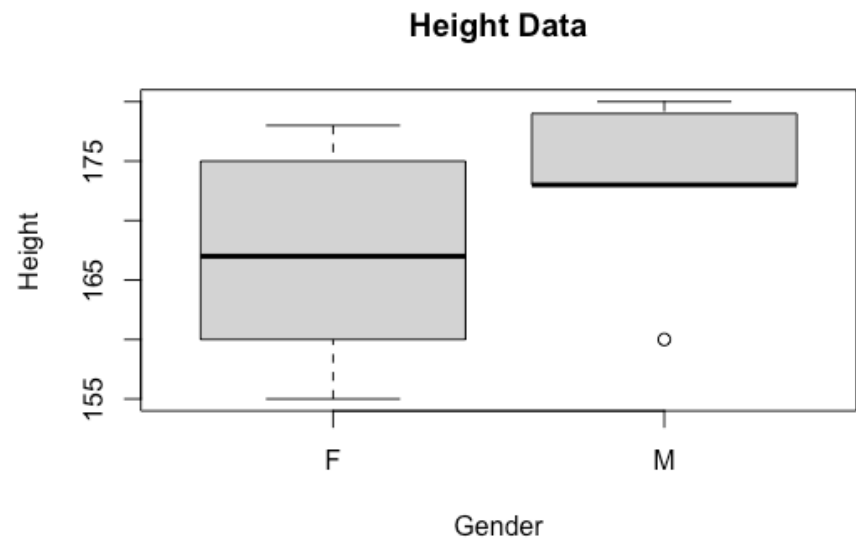


Figure shows that on an average height of male students is more than the female student.

- **Does this result apply in general for the population?**

For answering this question,

- ✓ We need to find the probability of occurrence of such sample data.
- ✓ Need to determine the probability, therefore, **Inferential analysis**

Inferential Analysis:

- Inferential analysis is performed to answer the question that
 - what is the probability of that the results obtained from an analysis can be applied to **the entire population**.

We can perform many different types of statistical tests on data.

Test	Why Performed?
Univariate Analysis: Z-Test or T-test	To determine, if the computed value of mean of a sample can be applicable for the population and related confidence interval.
Bivariate Chi-square test	To test the relationship between two categorical variables or groups of data
Two sample T-Test	To test the difference between the means of two variables or groups of data
One way ANOVA	To test the difference in mean of more than two variables or groups of data
F-Test	It can be used to determine the equality of variance of two or more groups of data
Correlation analysis	Determines the strength of relationship between two variables
Regression analysis	Examines the dependence of one variable over a set of independent variables
Decision Trees	Supervised learning used for classification
Clustering	Non-supervised Learning

Some tools for data analysis

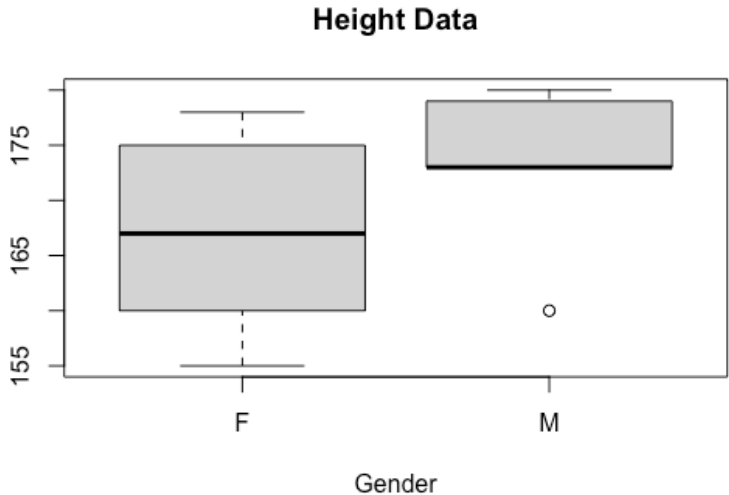
An example explains the importance of Inferential analysis.

Enrolment Number	Gender	Height
S20200001	F	155
S20200002	F	160
S20200003	M	179
S20200004	F	175
S20200005	M	173
S20200006	M	160
S20200007	M	180
S20200008	F	178
S20200009	F	167
S20200010	M	173

Sample Height Data

In order to infer, if there is a difference between the height of two groups (Male and Female Students), a two-sample t-test was run on the data.

t-Test (two tail): Assuming Unequal Variances		
	Female	Male
Mean	167	173
Variance	94.5	63.5
Observations	5	5
Computed t-value	-1.07	
p-value	0.32	
Critical t-value	2.30	



T-test shows that

- ✓ The **mean height** of the female students is 167 cm, whereas for the male students it is 173 cm.
- ✓ The **variance** of female candidates is 94.5, whereas for male candidate it is 63.5.
- ✓ Each group is interpreted on the basis of 5 observations.
- ✓ The computed t-value is -1.07 and p-value is 0.32.
- ✓ As the p-value is **greater than 0.05**, therefore
 - ✓ We cannot conclude that the average male student height is different from average female student height in the entire world population.

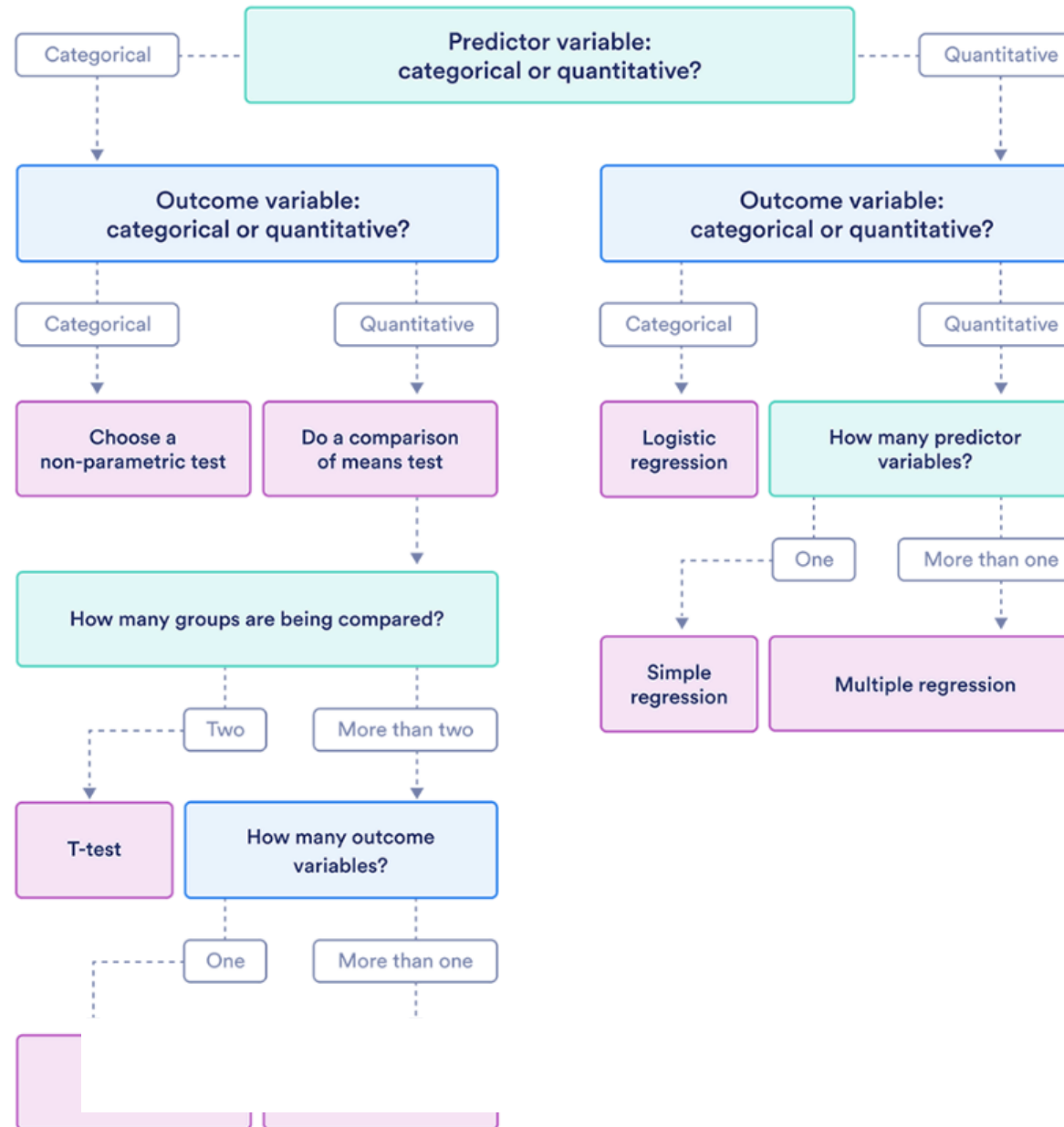
Predictive Analysis:

- The prescriptive analysis aims to
 - take predictions one step forward and
 - suggest solutions to present and future issues.

Statistical Test: Basics

- **Statistical tests** are used in **hypothesis testing**, which can be used to:
 - determine whether a **predictor variable** has a statistically significant relationship with an **outcome variable**.
 - estimate the difference between two or more groups.
- Statistical tests assume a **null hypothesis** of **no relationship** or **no difference** between groups.
 - Then, they determine whether the observed data fall outside of the range of values predicted by the null hypothesis.
- If we already know what **types of variables** we are dealing with, then
 - we can choose the right statistical test for our data. (See the flowchart)

Flowchart: choosing a statistical test



Understanding Statistical Hypothesis Testing

In general, statistical analysis is mainly used in the two situations:

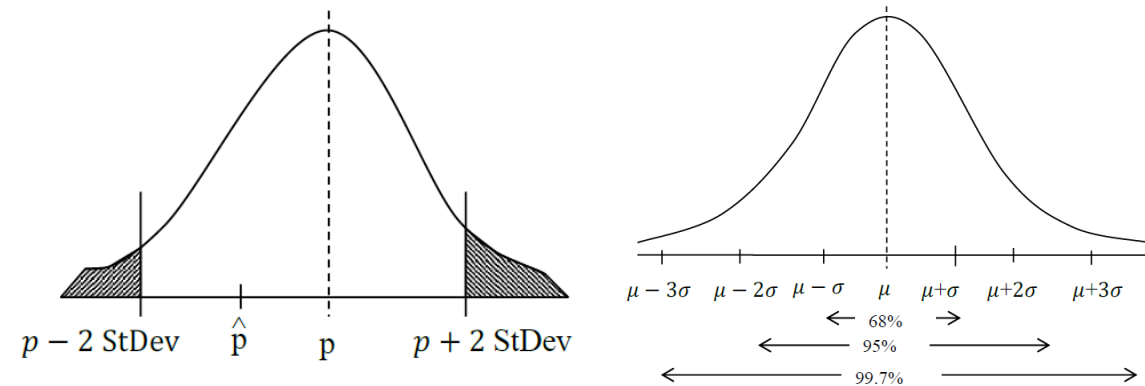
- **S1:** To determine if students of class 12 plays some sport, a sample random survey collected the data from 1000 students. Of these 405 students, stated that they play some sport. Using this information, can you infer that students of class 12 give less importance to sports?
 - ✓ Such a decision would require you to estimate the population parameters.
- **S2:** In order to study the effect of sports on the performance of class 12th marks, a study was performed. It performed random sampling and collected the data of 1000 students, which included information of Percentage of marks obtained by the student and hours spent by the student in sports per week during class 12th.
 - ✓ This kind of decision can be made through hypothesis testing.

S1: Estimation of Parameters of the Population

- One of the simplest ways to estimate the parameter value as a **point estimation**.
- Key characteristics of this estimate should be that it should be unbiased, such as
 - mean or median that lies towards the center of the data; and
 - should have small standard deviation, as far as possible.
- For example, a point estimate for situation S1 above would be that 40.5% students play some sports.
- This point estimate, however, may not be precise and may have some margin of error.
- Therefore, a better estimation would be to define an interval that contains the value of the parameter of the population.
- This interval, called confidence interval, includes the point estimate along with possible margin of error.
- The probability that the chosen confidence interval contains the population parameter is normally chosen as 0.95.
 - This probability is called the confidence level.
- Thus, we can state with 95% confidence that the confidence interval contains a parameter.

Is the value of confidence level as 0.95 arbitrary?

- As we know that sampling distribution for computing proportion is normal if the sample size (n) is large.
- Therefore, to answer the question asked above, we may study the Figure showing the probability distribution of sampling distribution.



Since we have selected a confidence level of 95%, we are expecting that proportion of the sample ($\hat{p}$) can be in the interval—

(population proportion (p) $- 2 \times$ (Standard Deviation)) to
(population proportion (p) $+ 2 \times$ (Standard Deviation))

- ✓ The probability of occurrence of $\hat{p}$ in this interval is 95%
- ✓ Therefore, the confidence level is 95%.

S1: Estimation of Parameters of the Population...

- Note that we do not know the value of p that is what we are estimating, therefore, we would be computing $\hat{p}$.
- We may observe in the Figure, that the value of p will be in the interval $(\hat{p} - 2 \cdot (\text{Standard Deviation}))$ to $(\hat{p} + 2 \cdot (\text{Standard Deviation}))$.

- However, as we are estimating the value of p , therefore, we cannot compute the exact value of standard deviation.
- Rather, we can compute standard error, which is computed by estimating the standard deviation using the sample proportion ($\hat{p}$), by using the following formula:

$$\text{Standard Error}(StErr) = \sqrt{\frac{\hat{p} \times (1 - \hat{p})}{n}}$$

- Therefore, the confidence interval is estimated as $(\hat{p} - 2 \cdot StErr)$ to $(\hat{p} + 2 \cdot StErr)$.
- In general, for a specific confidence level, we can specify a specific z-score instead of 2.
- Therefore, the confidence interval, for large n , is: $(\hat{p} - z \cdot StErr)$ to $(\hat{p} + z \cdot StErr)$
- In practice, we may use confidence level of 90% or 95% and 99%.
- The z-score used for these confidence levels are 1.65, 1.96 (not 2) and 2.58 respectively.

Example: Consider the statement S1 of this section and estimate the confidence interval for the given data.

- For the sample the probability that class 12th students play some sport is:

$$\hat{p} = 405/1000 = 0.405$$

- The sample size (n) = 1000

$$StErr = \sqrt{\frac{\hat{p} \times (1 - \hat{p})}{n}} = \sqrt{\frac{0.405 \times (1 - 0.405)}{1000}} = 0.016$$

Therefore, the Confidence Interval for the confidence level 95% would be:

$$\begin{aligned} &= (0.405 - 1.96 \cdot 0.016) \text{ to } (0.405 + 1.96 \cdot 0.016) \\ &= 0.374 \text{ to } 0.436 \end{aligned}$$

Therefore, with a confidence of 95%, we can state that the students of class 12th, who plays some sport is in the range 37.4% to 43.6%.

How can we reduce the size of this interval?

- We may observe that $StErr$ is inversely dependent on the square root of the sample size.
- Therefore, we may have to increase the sample size to approx 4 times to reduce the standard error to approx half.

S1: Estimation of Parameters of the Population...Example

- Example: The following table lists the height of a sample of 100 students of class 12 in centimetres. Estimate the average height of students of class 12.

170	164	168	149	157	148	156	164	168	160
149	171	172	159	152	143	171	163	180	158
167	168	156	170	167	148	169	179	149	171
164	159	169	175	172	173	158	160	176	173
159	160	162	169	168	164	165	146	156	170
163	166	150	165	152	166	151	157	163	189
176	185	153	181	163	167	155	151	182	165
189	168	169	180	158	149	164	171	189	192
171	156	163	170	186	187	165	177	175	165
167	185	164	156	143	172	162	161	185	174

The sample mean and sample standard deviation is computed and is shown below:

Sample Mean ($\bar{x}$) = 166; Standard Deviation of sample (s) = 11

Therefore, the estimated height confidence interval of the mean height of the students of class 12th can be computed as:

- Mean height ($\bar{x}$) = 166
- The sample size (n) = 100

$$\text{Standard Error in Sample Mean} = \frac{11}{\sqrt{100}} = 1.1$$

The Confidence Interval for the confidence level 95% would be:
(166 – 1.96 * 1.1) to (166 + 1.96 * 1.1) = 163.8 to 168.2

Thus, with a confidence of 95%, we can state that average height of class 12th students is in between 163.8 to 168.2 cm.

- We may note that in this example, we may use t-distribution for means, as we have used sample's standard deviation rather than population standard deviation.
- The t-distribution of means is slightly more restrictive than z distribution.
- The t -value is computed in the context of sampling distribution by the following equation:

$$t = \frac{(\bar{x} - \mu)}{s/\sqrt{n}}$$

S2: Significance Testing of Statistical Hypothesis

- A number of experimental studies are conducted in statistics, with the objective to infer, if the data support a hypothesis or not.
- The significance testing may involve the following phases:

1. Testing Pre-condition on Data:

- Prior to performing the test of significance, we should check the pre-conditions on the test.
- Most of the statistical test require 1) random sampling, 2) large size of data for each possible category being tested and 3) normal distribution of the population.

2. Making the statistical Hypothesis:

- We make statistical hypothesis after the parameters of the population.
- There are two basic hypothesis in statistical testing –
1) the **Null Hypothesis** and 2) the **Alternative Hypothesis**.
- **Null Hypothesis:** Null hypothesis either defines a particular value for the parameter or specifies there is no difference or no change in the specified parameters. It is represented as H_0 .
- **Alternative Hypothesis:** Alternative hypothesis specifies the values or difference in parameter values. It is represented as either H_1 or H_a . We use the convention H_a .

For example-

H_0 : There is no effect of hours of study on the mark's percentage of 12th class.

H_a : The marks of class 12th improves with the hours of study of the student.

- Note that the hypothesis above is one sided, as our assumption is that the marks would increase with hours of study.
- The second one sided hypothesis may relate to decrease in marks with hours of study.
- However, most of the cases the hypothesis will be two sided, which just claims that a variable will cause difference in the second.
- **For example**, two-sided hypothesis for statement S2 would be hours of study of students makes a difference (it may either increase or decrease) the marks of students of class 12th.
- In general, one-sided tests are called **one tailed tests** and **two-sided tests** are called two tailed tests.

In general, we may have alternative hypothesis which may be two way (increase or decrease; less or more etc.).

S2: Significance Testing of Statistical Hypothesis...

3. Perform the desired statistical analysis: (How to select?)

- Next, we perform the exploratory analysis and produce a number of charts to explore the nature of the data.
- This is followed by performing a significance statistical test like chi-square, independent sample t-test, ANOVA, nonparametric tests etc., which is decided on the basis of **size of the sample, type and characteristics of data.**
- These tests generate assumes the null hypothesis to be True.
- A test may generate parameter values based on sample and the probability called p-value, which is an evidence against the null hypothesis.
- This is shown in Figure

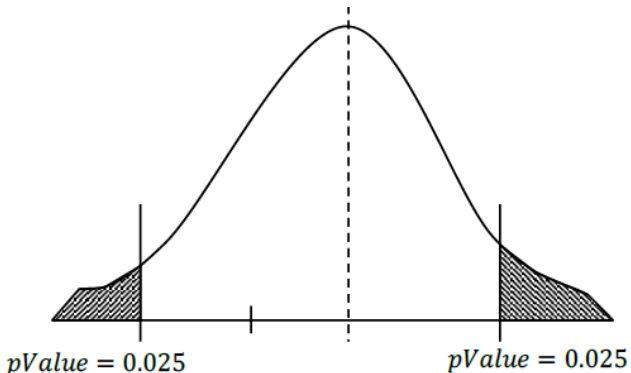


Figure : p-value of test statistics

4. Analysing the results:

- In this step, we should analyse our results.
- As stated, we must not just draw our conclusion based on statistics, but support it with analytical reasoning.

Example: We demonstrate the problem of finding a relationship between study hours and Marks percentage (as in S2),

- however, by using only sample data of 10 students (it is hypothetical data and just used for the illustration purpose), which is given as follows:

Weekly Study Hours (<i>wsh</i>)	96	92	63	76	89	80	56	70	61	81
Marks Percentage (<i>mp</i>)	21	19	7	11	16	17	4	9	7	18

In order to find such a relationship, we may like to perform basic exploratory analysis.

- In this case, let us make a scatter plot between the two variables, taking *wsh* as an independent variable and *mp* as a dependent variable.
- This scatter plot is in Figure

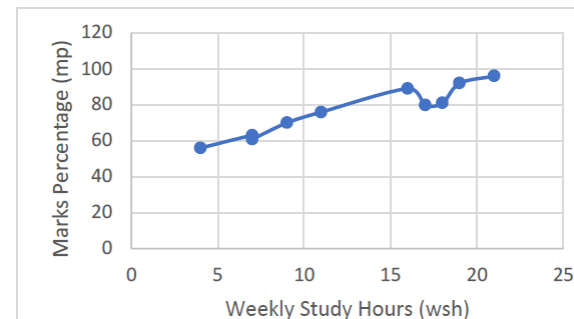


Figure suggests that the two variables may be associated. But how to determine the strength of this association?

In statistics, we use **Correlation**, which may be used to determine the strength of linear association between two quantitative variables.

Example using Correlation and Regression

As stated, correlation is used to determine the strength of linear association.

- But how the correlation is measured?
- Consider two quantitative variables x and y , and a set of n pairs of values of these variables (for example, the wsh and mp values as shown in example),
- we can compute a correlation coefficient, denoted by r using the following equation:

$$r_{xy} = \frac{\sum_{i=1}^n \left(\frac{(x-\bar{x})}{s_x} \right) \times \left(\frac{(y-\bar{y})}{s_y} \right)}{(n-1)}$$

The following are the characteristics of the correlation coefficient (r):

- The value of r lies between +1 and -1.
- A positive value of r means that value of y increases with increase in value of x and the value of y decreases with decrease in value of x .
- A negative value of r means that value of y increases with decrease in value of x and the value of y decreases with increase in value of x .
- If the value of r is closer to +1 or -1, then it indicates that association is a strong linear association.
- Simple scaling one of the variable does not change the correlation.
- Correlation does not specify the dependent and independent variables.
- Please remember a correlation does not mean cause. We have to establish it with reasoning.

- The data of Example shows a positive correlation.
- It can be computed as follows:

Mean of wsh = 12.9; Standard Deviation of wsh (Sample) = 5.98980616

Mean of mp = 76.4; Standard Deviation of mp (Sample) = 13.7210301

$$r_{wsh,mp} = \frac{8.61944034}{(10-1)} = 0.95771559$$

- As the linear correlation between wsh and mp variables is strong, therefore, we may like to find a line, called linear regression line, that may describe this association.
- The accuracy of regression line, in general, is better for higher correlation between the variables.

Single Linear Regression:

- A single linear regression predicts
 - a response variable or dependent variable (say y) using one
 - explanatory variable or independent variable (say x).
- The equation of single linear regression can be defined by using the following equation:

$$y_{predicted} = a + bx$$

Single Linear Regression:

$$y_{predicted} = a + bx$$

- Here,
 - $y_{predicted}$ is the predicted value of response variable (y),
 - x is the explanatory variable,
 - a is the intercept with respect to y and
 - b is called the slope of the regression line.
- In general, when we fit a linear regression line to a set of data, there will be certain difference between the $y_{predicted}$ and the observed value of data (say $y_{observed}$).
- This difference between the observed value and the predicted value, that is $(y_{observed} - y_{predicted})$, is called the residual.
- One of the most used method of finding the regression line is the method of least square, which minimizes the sum of squares of these residuals.
- The following equations can be used for computing residual:

$$Residual = y_{observed} - y_{predicted}$$

- The objective of least square method in regression is to minimise the sum of squares of the residual of all the n observed values.
- This sum is given in the following equation:

$$SumOfResidualSquares = \sum_{i=1}^n (y_{observed} - y_{predicted})^2$$

- Another important issue with regression model is to determine the predictive power of the model,
 - which is computed using the square of the correlation (r^2).
- The value of r^2 can be computed as follows:

- In case, we are not using regression, then we can predict the value of y using the mean.
 - ✓ In such a case, the difference in predicted value and observed value would be given by the following equation:

$$ErrorUsingMean = y_{observed} - \bar{y}$$

- The total of sum of square of this error can be computed using the following equation:

$$TotalSumOfSquare = \sum_{i=1}^n (y_{observed} - \bar{y})^2$$

- Thus, use of regression results helps in reducing the error.
- The proportion r^2 is actually the predictive power of the regression and is represented using the following equation:

$$r^2 = \frac{\sum_{i=1}^n (y_{observed} - \bar{y})^2 - \sum_{i=1}^n (y_{observed} - y_{predicted})^2}{\sum_{i=1}^n (y_{observed} - \bar{y})^2}$$

As stated earlier, r^2 can also be computed by squaring the value of r .

Linear Regression Model Evaluation...

- **Adjusted R-squared — selection criterion**
- The main difference between **adjusted R-squared** and R-square is that
 - **R-squared** describes the amount of variance of the dependent variable represented by every single independent variable, while
 - **adjusted R-squared** measures variation explained by only the independent variables that actually affect the dependent variable.

$$R^2_{adjusted} = \left[\frac{(1-R^2)(n-1)}{n-k-1} \right]$$

- ✓ *n is the number of data points while k is the number of variables in our model, excluding the constant.*
- R^2 tends to increase with an increase in the number of independent variables. **This could be misleading.**
- Thus, the adjusted R-squared penalizes the model for adding furthermore independent variables (k in the equation) that do not fit the model.

Single Linear Regression: Example

On performing regressing analysis on the observed data of Example, the statistics as shown in the Figure is generated.

Regression Statistics	
Multiple R	0.9577
R Square	0.9172
Adjusted R Square	0.9069
Standard Error	4.1872
Observations	10.0000

- ✓ The term “Multiple R” in *Regression Statistics* defines the correlation between the dependent variable (say y) with the set of independent or explanatory variables in the regression model.
- ✓ Thus, multiple R is similar to correlation coefficient (r), expect that it is used when multiple regression is used.

ANOVA				
	<i>df</i>	<i>SS</i>	<i>F</i>	<i>Significance F</i>
Regression	1.0000	1554.1361	88.6407	0.0000
Residual	8.0000	140.2639		
Total	9.0000	1694.4000		

- The *Significance F* in ANOVA is 0, therefore, you can reject the Null hypothesis H_{0A} and determine that the this model can predict the value of y . Please note high F value supports this observation.
- We can compute the sum of squares (SS) using Equations
- The degree of freedom in the context of statistics is the number of data items required to compute the desired statistics.

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
<i>Intercept</i>	48.0991	3.2847	14.6435	0.0000
Weekly Study Hours (<i>wsh</i>)	2.1939	0.2330	9.4149	0.0000

- The p-value related to *intercept* and *wsh* are almost 0, therefore, you can reject the Null hypothesis H_{0I} and H_{0wsh} .
- The regression line has the equation:
$$mp_{predicted} = 48.0991 + 2.1939 \times wsh$$

Single Linear Regression: Example

- Figure 19 shows the regression line for the data of Example.
- We may observe that residuals is the vertical difference between the Marks Percentage and Predicted marks percentage. These residuals are shown in Figure 20.

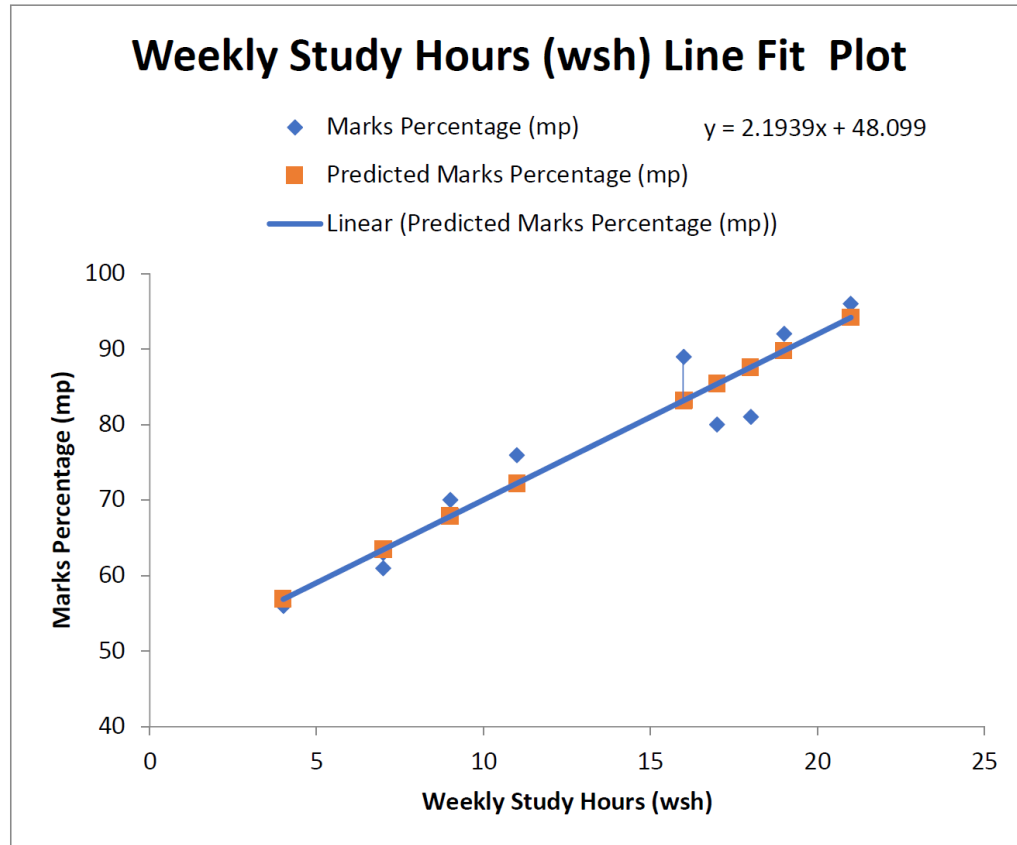


Figure 19: The Regression Line

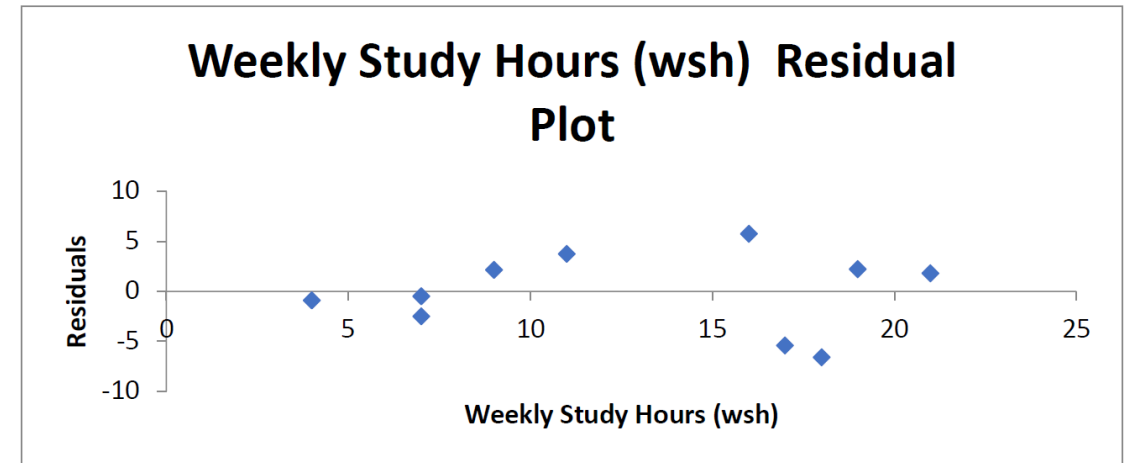


Figure 20: The Residual Plot

2.4: Interpreting Results of regression

```
Residuals:
    Min       1Q   Median       3Q      Max
-5.012 -1.713  0.313  1.725  3.416

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -33.1629     7.4783  -4.435  0.00218 **
x              0.6379     0.0486  13.125 1.08e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.807 on 8 degrees of freedom
Multiple R-squared:  0.9556,    Adjusted R-squared:  0.9501
F-statistic: 172.3 on 1 and 8 DF,  p-value: 1.08e-06
```

- The results of regression (as presented by R) includes the following:
 1. Five-point summary (Minimum, First Quartile, Median, Third Quartile, and Maximum). This shows the spread of the residual.
 - We may observe that about 50% of residuals are in the range -1.713 to +1.725, which shows a good model fit.
 2. The t value and Pr values for the intercept (that is b in the equation $y = ax + b$) and x (that is a in the equation $y = ax + b$).
 3. The F-statistics is very high with a low p-value, indicating statistical difference between group means.

Solving a regression problem:

- Solving a regression problem involves obtaining a relation between
✓ the response Y and the predictors
- i.e. determine a function $f(X_1, X_2, \dots, X_p)$ such that $Y = f(X) + \epsilon$,
✓ based on the given set D of patterns.

where ϵ is called a random error such that $E(\epsilon) = 0$

	Advertising Budget on			Sales (Y)
	TV (X_1)	Radio (X_2)	News Paper (X_3)	
x_1	x_{11}	x_{12}	x_{13}	y_1
x_2	x_{21}	x_{22}	x_{23}	y_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_i	x_{i1}	x_{i2}	x_{i3}	y_i
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_N	x_{N1}	x_{N2}	x_{N3}	y_N

➤ In Multiple Linear Regression $f(X)$ is assumed to be in the form:

(Model) $f(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$ (1)

$$y \approx \hat{y} = f(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$$

Goal: Determine β so as to satisfy the following N equations:

$$\left. \begin{aligned} y_1 &= \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \dots + \beta_p x_{1p} \\ y_2 &= \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \dots + \beta_p x_{2p} \\ &\vdots \\ y_i &= \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_j x_{ij} + \dots + \beta_p x_{ip} \\ &\vdots \\ y_N &= \beta_0 + \beta_1 x_{N1} + \beta_2 x_{N2} + \dots + \beta_j x_{Nj} + \dots + \beta_p x_{Np} \end{aligned} \right\} \begin{array}{l} N \text{ equations and} \\ (p + 1) \text{ unknowns} \end{array}$$

Fact from Linear Algebra:

$$\left. \begin{array}{l} y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \cdots + \beta_p x_{1p} \\ y_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \cdots + \beta_p x_{2p} \\ \vdots \\ y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_j x_{ij} + \cdots + \beta_p x_{ip} \\ \vdots \\ y_N = \beta_0 + \beta_1 x_{N1} + \beta_2 x_{N2} + \cdots + \beta_j x_{Nj} + \cdots + \beta_p x_{Np} \end{array} \right\} \begin{array}{l} \text{N equations and} \\ (p+1) \text{ unknowns} \end{array}$$

- The above system may be written as $X\beta = y$, where X is an $N \times (p+1)$ matrix and y is an $N \times 1$ vector.
- For an over determined system, these equations have a solution only if y is a linear combination of columns of X .
- For most choices of y , however, there is no vector β for which $X\beta = y$.

So, we seek a β for which the norm of $r = y - X\beta$ (called the residual) is as small as possible.

We have $\hat{y}_i = f(x_i)$, where $\hat{y}_i$ denotes the computed value from the model for input x_i

Residual (error) $= y_i - \hat{y}_i$

Residual Sum of Squares (RSS) is defined as

$$RSS(\beta) = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

New Goal: Determine β so as to minimise $RSS(\beta)$ where,

$$RSS(\beta) = \sum_{i=1}^N [y_i - (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip})]^2$$

is a function of $f(\beta_0, \beta_1, \dots, \beta_p)$

We need optimization!

Positive definite matrix (Definition): A symmetric matrix A of order $n \times n$ is said to be positive definite if $x^T A x > 0$ for all $x \neq 0, x \in \mathbb{R}^n$.

Theorem: If A is +ve definite then all its Eigen values are +ve.

✓ $RSS(\beta)$ is minimized for some $\beta = \tilde{\beta}$

✓ if and only if $\nabla f|_{\beta=\tilde{\beta}} = 0$ and Hessian of f at $\beta = \tilde{\beta}$ is +ve definite.

$$\nabla f = \begin{pmatrix} \frac{\partial f}{\partial \beta_0} \\ \frac{\partial f}{\partial \beta_1} \\ \frac{\partial f}{\partial \beta_2} \\ \vdots \\ \frac{\partial f}{\partial \beta_p} \end{pmatrix}, \text{ and Hessian of } f = \begin{bmatrix} \frac{\partial^2 f}{\partial \beta_0^2} & \frac{\partial^2 f}{\partial \beta_0 \partial \beta_1} & \frac{\partial^2 f}{\partial \beta_0 \partial \beta_2} & \cdots & \frac{\partial^2 f}{\partial \beta_0 \partial \beta_p} \\ \frac{\partial^2 f}{\partial \beta_1 \partial \beta_0} & \frac{\partial^2 f}{\partial \beta_1^2} & \frac{\partial^2 f}{\partial \beta_1 \partial \beta_2} & \cdots & \frac{\partial^2 f}{\partial \beta_1 \partial \beta_p} \\ \frac{\partial^2 f}{\partial \beta_2 \partial \beta_0} & \frac{\partial^2 f}{\partial \beta_2 \partial \beta_1} & \frac{\partial^2 f}{\partial \beta_2^2} & \cdots & \frac{\partial^2 f}{\partial \beta_2 \partial \beta_p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial \beta_p \partial \beta_0} & \frac{\partial^2 f}{\partial \beta_p \partial \beta_1} & \frac{\partial^2 f}{\partial \beta_p \partial \beta_2} & \cdots & \frac{\partial^2 f}{\partial \beta_p^2} \end{bmatrix}$$

Solved Example: For which values of x_1 and x_2 the given function $f(x_1, x_2) = x_1^2 + x_1x_2 + x_2^2 - 2x_1$ is minimized.

Solution:

Since, gradient of $f = \nabla f = \begin{pmatrix} \frac{\partial f}{\partial x_1} = 2x_1 + x_2 - 2 \\ \frac{\partial f}{\partial x_2} = x_1 + 2x_2 \end{pmatrix}$

On equating the gradient to zero that is $\nabla f = 0$,

Hence, we get

$$2x_1 + x_2 - 2 = 0 \quad (\text{i})$$

$$x_1 + 2x_2 = 0 \quad (\text{ii})$$

from equation (ii)

$$\text{we have } x_1 = -2x_2 \quad (\text{iii})$$

on substituting the value of x_1 in equation (i)

we have

$$2(-2x_2) + x_2 - 2 = 0 \Rightarrow -4x_2 + x_2 = 2 \\ \Rightarrow -3x_2 = 2 \Rightarrow x_2 = -\frac{2}{3}$$

on substituting the value of x_2 in equation (iii)

$$\text{we have } x_1 = \frac{4}{3}$$

$$\text{Now, } H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

To find eigen values,

$$\begin{aligned} |H_f - \lambda I| &= 0 \\ \Rightarrow \left| \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| &= 0 \\ \Rightarrow \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} &= 0 \\ \Rightarrow (2-\lambda)^2 - 1 &= 0 \\ \Rightarrow 2-\lambda = +1 \text{ and } 2-\lambda = -1 \\ \Rightarrow \lambda = 1 \quad \text{and } \lambda = 3 \end{aligned}$$

Hence, H_f is positive definite.

Therefore the given function

$f(x_1, x_2) = x_1^2 + x_1x_2 + x_2^2 - 2x_1$ is minimized for

$$x_1 = \frac{4}{3} \text{ and } x_2 = -\frac{2}{3}.$$

Limitation of Multiple Linear Regression

1. If the data are not linearly related to the response then the prediction accuracy of the Multiple Linear Regression is low.
2. If some features are highly correlated then the matrix X may become computationally singular and hence $\hat{\beta}$ cannot be computed. If two features are highly correlated then remove one of them but not both.
3. When p is large linear regression fit does not predict well even when the data is not non-linear.

To overcome these limitations one of the approaches is shrinking the coefficient (β 's) of the model or setting some of the β 's equal to zero.

Predictive Analysis:

- A detailed discussion on these topics is beyond the scope of this Unit.

Understanding AI/ML

- Machine Learning is a branch of Artificial intelligence that focuses on the development of algorithms and statistical models
 - that can learn from and make predictions on data.
- **Supervised algorithm** is a type of machine learning where the algorithm learns from labeled data.
 - Labeled data means the dataset whose respective target value is already known.
- Supervised learning has two types:
 - **Regression:**
 - It predicts the **continuous output variables** based on the independent input variable.
 - like the prediction of house prices based on different parameters like house age, distance from the main road, location, area, etc.
 - **Classification:**
 - It predicts the **class of the dataset** based on the independent input variable.
 - Class is the categorical or discrete values. like the image of an animal is a cat or dog?
- **Unsupervised Machine Learning** is the technique where models are not provided with the labeled data and
- They have to find the patterns and structure in the data to know about the data.
 - **Clustering and Association** algorithms are a part of Unsupervised ML.

1. How to measure the effectiveness of AI/ML models?

- When we get the data, (after data cleaning, pre-processing, and wrangling), the first step we do is
 - to feed it to an outstanding model and
 - get output in probabilities.
- But, How can we measure the effectiveness of our model?
 - Better the effectiveness, better the performance.
- **Supervised:**
 - **Regression** model's performance:
 - MAE, MSE, RMSE, R-squared, and Adjusted R-squared.
 - **Classification** model's performance:
 - Confusion Matrix and AUROC (Area Under the Receiver Operating Characteristics)
- **Unsupervised:**
 - **Clustering** model's performance:
 - Silhouette Coefficient
 - Calinski-Harabasz Index
 - Davies-Bouldin Index
 - **Association** model's performance: To measure the associations between thousands of data items, there are several metrics. Some of them are:
 - Support
 - Confidence
 - Lift

Types of Errors in Hypothesis Testing

- We either Reject the Null hypothesis and accept alternative hypothesis based on
 - the computed probability or p-value;
 - or you fail to Reject the Null hypothesis.
- The decisions in such hypothesis testing would be:

- ✓ We reject Null hypothesis for a confidence interval 95% based on the p-value, which lies in the shaded portion, that is $p\text{-value} < 0.05$ for two tailed hypothesis (that is both the shaded portions in Figure, each area of probability 0.025).
 - Note that in case of one tailed test, we would consider only one shaded area, therefore, we would be considering $p\text{-value} < 0.05$ in only one of the two shaded areas.

- ✓ We fail to reject the Null hypothesis for confidence interval 95%, when $p\text{-value} > 0.05$.

- The two decisions as stated could be incorrect, as we are considering a confidence interval of 95%.
- The following Figure shows this situation.

<i>The Actual Scenario</i>	<i>Final Decision</i>	
	<i>H_0 is Rejected, that is, you have accepted the Alternative hypothesis</i>	<i>You fail to reject H_0, as you do not have enough evidence to accept the Alternative hypothesis</i>
H_0 is True	This is called a TYPE-I error	You have arrived at a correct decision
H_0 is False	You have arrived at a correct decision	This is called a TYPE-II error

For example, assume that a medicine is tested for a disease and this medicine is NOT a cure of the disease.

- We would make the following hypotheses:

H_0 : The medicine has no effect for the disease

H_a : The medicine improves the condition of patient.

- However, if the data is such that for a confidence interval of 95% the p-value is computed to be less than 0.05, then we will reject the null hypothesis, which is Type-I error.
- The chances of Type-I errors for this confidence interval is 5%.
- This error would mean that the medicine will get approval, even though it has no effect on curing the disease.

- However, now assume that a medicine is tested for a disease and this medicine is a cure of the disease.
- Hypotheses still remains the same, as above.
- However, if the data is such that for a confidence interval of 95% the p-value is computed to be more than 0.05, then we will not be able to reject the null hypothesis, which is Type-II error.
- This error would mean that a medicine which can cure the disease will not be accepted.

More Example on Classification and Regression Problems

- KNN is used to solve both Classification and Regression Problems.

KNN (K - Nearest Neighbour) for Classification Problem:

Given: $D = \{(x_i, y_i)\}_{i=1}^N$, $x_i \in \mathbb{R}^p$, $y_i \in \{1, 2, 3, \dots, M\}$, K ,
and an $x \in \mathbb{R}^p$, $x \neq x_i$

Goal: Determine the response $y \in \{1, 2, 3, \dots, M\}$ associated with $x \in \mathbb{R}^p$.

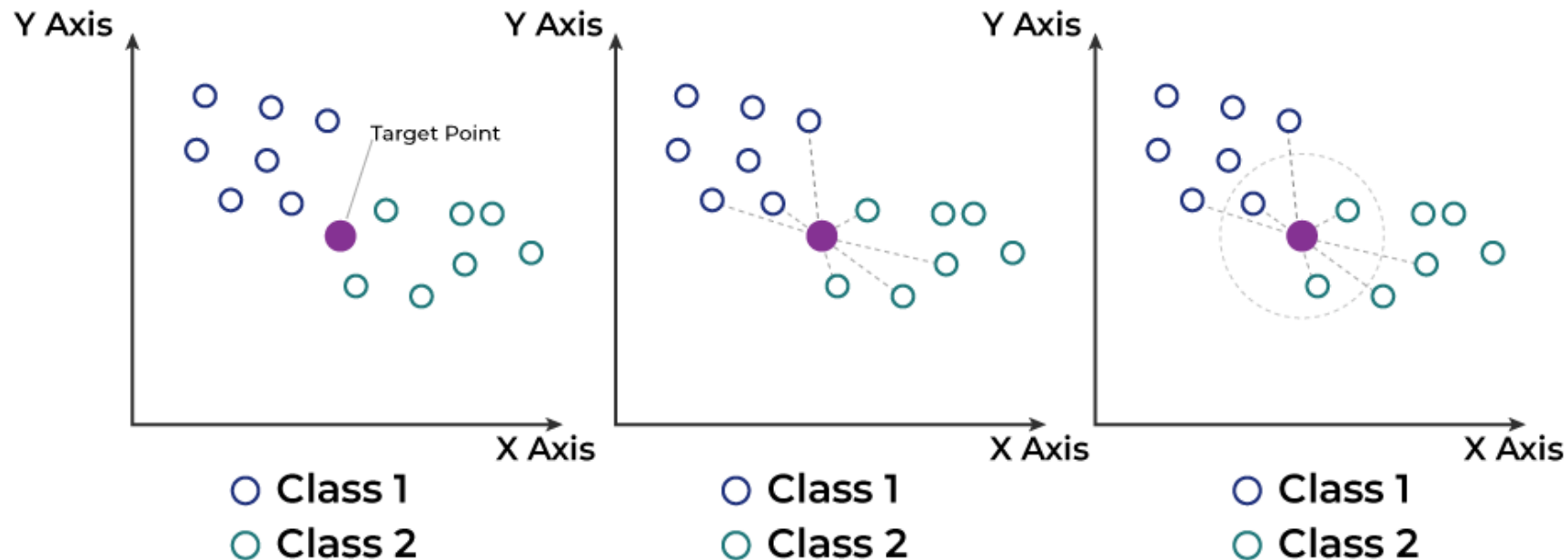
KNN for Regression Problem

Given: $D = \{(x_i, y_i)\}_{i=1}^N$, $x_i \in \mathbb{R}^p$, $y_i \in \mathbb{R}$, K ,
and an $x \in \mathbb{R}^p$, $x \neq x_i$

Goal: Determine the response $y \in \mathbb{R}$ associated with $x \in \mathbb{R}^p$.

Workings of KNN algorithm

- The K-Nearest Neighbors (KNN) algorithm operates on the principle of similarity, where
 - it predicts the **label** or **value** of a new data point by considering the labels or values of its K nearest neighbors in the training dataset.



Step-by-Step explanation of how KNN works:

Step 1: Selecting the optimal value of K

- K represents the number of nearest neighbors that needs to be considered while making prediction.

Step 2: Calculating distance

- To measure the similarity between target and training data points, Euclidean distance is used. Distance is calculated between each of the data points in the dataset and target point.

$$\text{distance}(x, X_i) = \sqrt{\sum_{j=1}^d (x_j - X_{i_j})^2}$$

Step 3: Finding Nearest Neighbors

- The k data points with the smallest distances to the target point are the nearest neighbors.

Step 4: Voting for Classification or Taking Average for Regression

- In the classification problem, the class labels of K-nearest neighbors are determined by performing majority voting. The class with the most occurrences among the neighbors becomes the predicted class for the target data point.
- In the regression problem, the class label is calculated by taking average of the target values of K nearest neighbors. The calculated average value becomes the predicted output for the target data point.

Example

We have data from the questionnaires survey (to ask people opinion) and objective testing with two attributes (acid durability and strength) to classify whether a special paper tissue is good or not. Here is four training samples

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Y = Classification
7	7	Bad
7	4	Bad
3	4	Good
1	4	Good

Now the factory produces a new paper tissue that pass laboratory test with $X1 = 3$ and $X2 = 7$. Without another expensive survey, can we guess what the classification of this new tissue is?

Solution: *1. Determine parameter K = number of nearest neighbors*

Suppose use K = 3

2. Calculate the distance between the query-instance and all the training samples

Coordinate of query instance is (3, 7), instead of calculating the distance we compute square distance which is faster to calculate (without square root)

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Square Distance to query instance (3, 7)
7	7	$(7-3)^2 + (7-7)^2 = 16$
7	4	$(7-3)^2 + (4-7)^2 = 25$
3	4	$(3-3)^2 + (4-7)^2 = 9$
1	4	$(1-3)^2 + (4-7)^2 = 13$

Solution Cont...:

3. *Sort the distance and determine nearest neighbors based on the K-th minimum distance*

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Square Distance to query instance (3, 7)	Rank minimum distance	Is it included in 3- Nearest neighbors?
7	7	$(7-3)^2 + (7-7)^2 = 16$	3	Yes
7	4	$(7-3)^2 + (4-7)^2 = 25$	4	No
3	4	$(3-3)^2 + (4-7)^2 = 9$	1	Yes
1	4	$(1-3)^2 + (4-7)^2 = 13$	2	Yes

Solution Cont...:

4. *Gather the category **Y** of the nearest neighbors.* Notice in the second row last column that the category of nearest neighbor (Y) is not included because the rank of this data is more than 3 (=K).

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Square Distance to query instance (3, 7)	Rank minimum distance	Is it included in 3-Nearest neighbors?	Y = Category of nearest Neighbor
7	7	$(7-3)^2 + (7-7)^2 = 16$	3	Yes	Bad
7	4	$(7-3)^2 + (4-7)^2 = 25$	4	No	-
3	4	$(3-3)^2 + (4-7)^2 = 9$	1	Yes	Good
1	4	$(1-3)^2 + (4-7)^2 = 13$	2	Yes	Good

5. *Use simple majority of the category of nearest neighbors as the prediction value of the query instance*

We have 2 good and 1 bad, since $2 > 1$ then we conclude that a new paper tissue that pass laboratory test with $X1 = 3$ and $X2 = 7$ is included in **Good** category.

Example 2: The table below provides a training data set containing six observations, three predictors X1, X2, X3 and one qualitative response variable Y.

Observation	X1	X2	X3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	-1	0	1	Green
6	1	1	1	Red

- Predict the value of Y when $X1 = 1$, $X2=5$, $X3 = 0$ using KNN with $k=3$.

Sol:

Observation	X1	X2	X3	Square distance to query instance (X1 = 1, X2=5, X3 = 0)	Rank Minimum Distance	Is it included in 3-NN	Y
1	0	3	0	$(0 - 1)^2 + (3 - 5)^2 + (0 - 0)^2$ $= 1 + 4 + 0 = 5$	1	Y	Red
2	2	0	0	$(2 - 1)^2 + (0 - 5)^2 + (0 - 0)^2$ $= 1 + 25 + 0 = 26$	5	N	-
3	0	1	3	$(0 - 1)^2 + (1 - 5)^2 + (3 - 0)^2$ $= 1 + 16 + 9 = 26$	4	N	-
4	0	1	2	$(0 - 1)^2 + (1 - 5)^2 + (2 - 0)^2$ $= 1 + 16 + 4 = 21$	3	Y	Green
5	-1	0	1	$(-1 - 1)^2 + (0 - 5)^2 + (1 - 0)^2$ $= 4 + 25 + 1 = 30$	6	N	-
6	1	1	1	$(1 - 1)^2 + (1 - 5)^2 + (1 - 0)^2$ $= 0 + 16 + 1 = 17$	2	Y	Red

- The predict value of Y when X1 = 1, X2=5, X3 = 0 using KNN with k=3 is **Red**.

Advantages of KNN

- It does not make any assumption concerning the distribution of data.
- So, the method is applicable to wide class of data.

Disadvantages of KNN

1. It provides limited insight into relationship between the predictors and the classes.
2. Here one needs the entire dataset whenever one wants to classify a new point x in R^p .
So, if N or p is large the computation is prohibitive.
3. The prediction is slow, because it requires comparing **distances to every point**.

Remarks:

1. The accuracy of the method depends upon the choice of K . So, one has to tune the value of K suitably.

Theorem: The expected error of k -NN rule converges to $1 + \sqrt{8/k}$ times the error of the Bayes classifier.

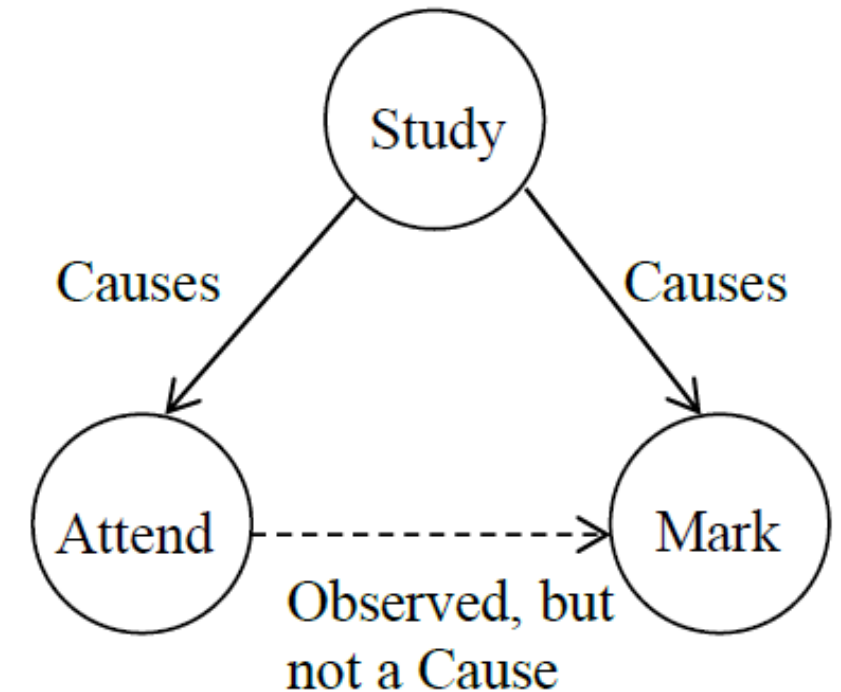
2. When the number of features p is large, there tends to be a deterioration in the performance of KNN.

Common Misconceptions of Data Analysis:

we discuss **three** misconception that can affect the result of a data science.

[1] Correlation analysis establishes relationship between two variables.

- For example, consider three variables, namely
 - attendance of student (attend),
 - marks obtained by student (marks) and
 - weekly hours spent by a student for the study (study).
- While analysing data, we found that there is a strong correlation between the variables **attend** and **marks**.
 - However, does it really mean that higher attendance causes students to obtain better marks?
- There is another possibility that both **study** and **marks**, as well as **study** and **attend** are correlated.
 - A motivated student may be spending higher number of hours at home, which may lead to better marks.
- Similarly, a motivated student who is putting a greater number of hours in his/her study may be attending to school regularly.
- Thus, the correlation between study to marks and study to attend results in a non-existing correlation attend to marks.
- This situation is shown in Figure



Correlation does not mean causation

[2] *Simpsons Paradox:*

- Simson paradox is an interesting situation, which sometimes leads to wrong interpretations.
- Consider two Universities, say University 1 and University 2 and the pass out data of these Universities:

University	Student Passed	Passed %	Student Failed	Failed %	Total
U1	4850	97%	150	3%	5000
U2	1960	98%	40	2%	2000

- As we may observe from the data as above,
 - ✓ the University U2 is performing better as far as passing percentage is concerned.
 - ✓ However, during a detailed data inspection,
 - it was noted that both the Universities were running Basic Adult Literacy Programme, which in general has a slightly poor result.
 - In addition, the data of the literacy Programme is to be compiled separately.
- Therefore, the be data for the University would be:

General Programmes:

University	Student Passed	Passed %	Student Failed	Failed %	Total
U1	1480	98.7%	20	3%	1500
U2	1480	98.7%	20	2%	1500

Adult Literacy Programme:

University	Student Passed	Passed %	Student Failed	Failed %	Total
U1	3370	96.3%	130	3.7%	3500
U2	480	96%	20	2%	500

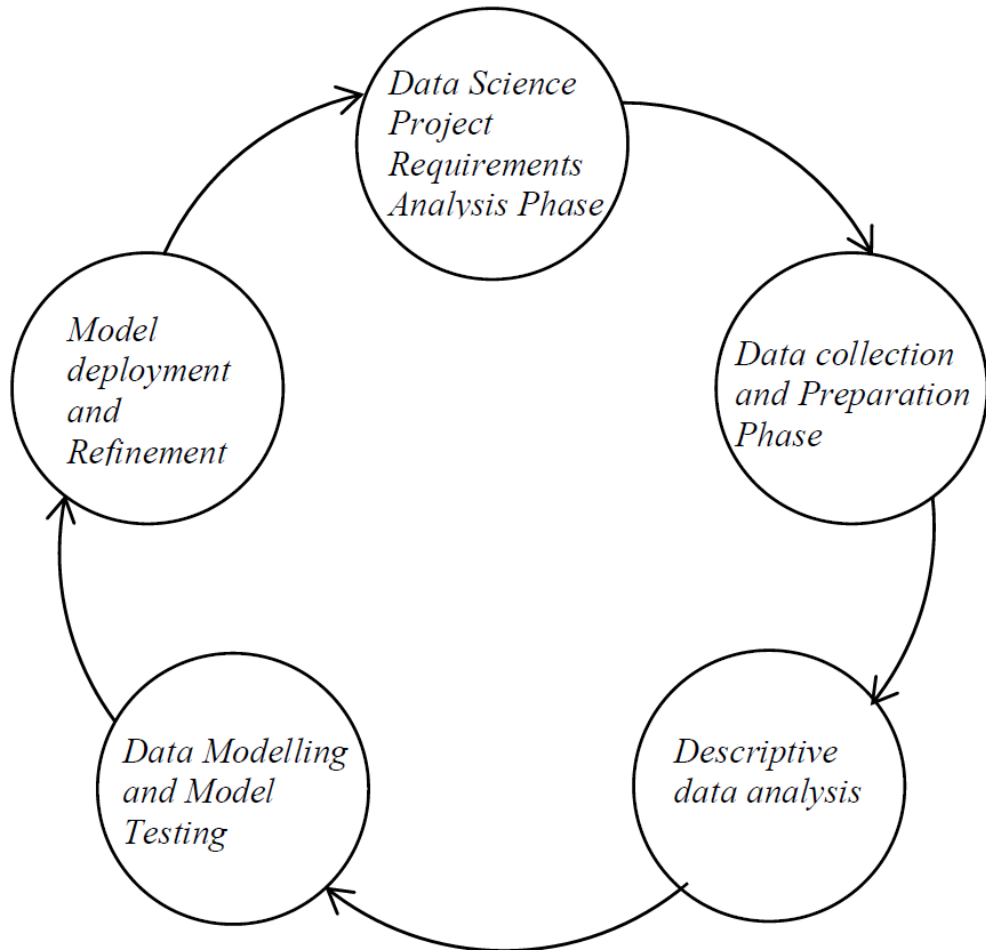
- We may observe that due to the additional grouping due to adult literacy programme,
 - ✓ the corrected data shows that U1 is performing better than U2,
 - ✓ as the pass out rate for General programme is same and pass out rate for Adult literacy programme is better from U1.
- We must note the changes in the percentages.
- This is the **Simpson's paradox**

[3] *Data Dredging:*

- It is extensive analysis of very large data sets.
- Such analysis results in generation of large number of data associations.
- Many of those associations may not be casual, thus,
 - requires further exploration through other techniques.
- Therefore, it is essential that every data association with large data set should be investigated further
 - before reporting them as conclusion of the study.

UNIT 1: INTRODUCTION TO DATA SCIENCE

- Basic Methods of Data Analysis
- Descriptive Analysis, Exploratory Analysis
- Inferential Analysis, Predictive Analysis
- Common Misconceptions of Data Analysis
- **Applications of Data Science**
- **Data Science Life cycle**



A sample Life cycle of Data Science Project

- Data Science is useful in analysing large data sets to produce useful information that can be used
 - for business development and
 - can help in decision making process.
- **Applications using Similarity analysis:**
 1. **Spam detection system:** This system classifies the emails into spam and non-spam categories.
 2. **Financial Fraud detection system:**
 - This is one of the major applications for online financial services.
 - Basic principle is once again to classify the transactions as safe or unsafe transactions based on various parameters of the transactions.
 3. **Recommendation of Products:**
 - Several e-commerce companies have the data of your buying patterns, information about your searches to their portal and other information about your account with them.
 - This information can be **clustered** into classes of buyers, which can be used to recommend various products for you.
- **Applications related to Web Searching**
- **Applications related to Healthcare System**
- **Applications related to Transport sector**